

Behavioural Responses to Targeted Housing Subsidies: Evidence from Ireland's Help to Buy Scheme

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Sciences Po Paris, School of Research
Department of Economics

Michelle Alicia Jordan

Under the Supervision of
Antoine Ferey

Jury Member: **Pierre-Philippe Combes**

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Michelle Alicia Jordan[†]

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Abstract

When a housing subsidy is targeted, restricted to a particular buyer group, dwelling type, or price range, it can distort behaviour along two distinct margins. First, at the eligibility boundary, there is an incentive to keep transactions just below the cut-off. Second, among buyers eligible for the subsidy, demand can shift towards the dwellings that qualify for it. This paper studies both responses in the context of Ireland’s Help to Buy scheme, a tax-based subsidy for first-time buyers of new-build dwellings valued at or below €500,000. Introduced in 2017 at 5% of the transaction-value, capped at €20,000, the scheme was enhanced in 2020 to 10%, capped at €30,000. Combining a developer-side notch model with evidence from administrative data, I find clear bunching just below the €500,000 threshold in the new-build market that strengthens monotonically with subsidy generosity. No such bunching exists in second-hand transactions or before the policy was implemented. The implied transaction-value response of the marginal developer who relocates below this threshold is €6,759 under the initial regime and €10,448 under the enhanced regime, around one-third of the maximum statutory relief in each regime. At the same time, under the initial regime, the first-time buyer share of new-build purchases rises by 5.11 percentage points relative to the second-hand market, and the new-build share of first-time buyer purchases increases by 2.48 percentage points relative to former owner-occupiers. Both estimates approximately double under the enhanced regime, in line with the doubling of the headline subsidy rate. The threshold distortion and the participation response are not independent: in a targeted scheme, the size of the threshold distortion depends on the share of demand that is sensitive to the loss of the subsidy at the threshold. More generous subsidies therefore strengthen both margins together, through the value of eligibility itself, and through the share of demand exposed to the cut-off. (**JEL:** H24, R31, R38, R21.)

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[†]michelle.jordan@sciencespo.fr.

1 Introduction

Housing purchase subsidies are a common policy instrument for promoting homeownership and supporting housing market activity, particularly amongst younger and liquidity-constrained households. Yet when eligibility depends on a transaction-value cut-off, the policy creates a discontinuous incentive to remain below that threshold, and when it is also restricted to a particular buyer group or dwelling type, it alters which buyers participate in the eligible market and what they purchase. A growing empirical literature documents threshold responses in housing markets (Agarwal, Hu and Lee, 2023; Uribe, 2022; Best and Kleven, 2018; Kopczuk and Munroe, 2015), and Hargaden (2017) shows that buyer-specific tax exemptions can generate allocative distortions across buyer types at a threshold. Less is known, however, about how these two margins behave together when a subsidy is targeted on a subset of demand within a defined market segment: whether the local threshold distortion is attenuated by the restricted coverage, whether the participation margin responds at all, and whether both respond in the same direction as subsidy generosity rises.

This paper studies Ireland’s Help to Buy (HTB) scheme, a tax-based subsidy for first-time buyers of new-build dwellings with a transaction-value at or below €500,000. Introduced in January 2017, the scheme initially provided relief equal to 5% of the transaction-value, capped at €20,000, and was enhanced in July 2020 to 10%, capped at €30,000. Second-hand properties have never been eligible. This setting is well suited to studying threshold and participation responses jointly: the policy creates a salient notch at €500,000 in the new-build market, leaves a natural ineligible comparison group in the second-hand market, and contains a discrete increase in subsidy generosity within the same institutional framework. Unlike much of the existing notch literature, which studies transaction taxes or price ceilings that apply to the full market, HTB applies only to a subset of buyers, so the size of the threshold distortion depends on the share of demand that is sensitive to losing eligibility at the cut-off.

To study both margins together, I propose a developer-side notch model that makes the link between both responses explicit, and combine it with three empirical strategies. First, using transaction-level data from the Property Price Register, I estimate bunching at the €500,000 threshold using a polynomial counterfactual. Second, I implement a distributional difference-in-differences comparing new-build and second-hand transactions across both policy regimes, locating the threshold response within the wider distribution and distinguishing a local distortion from diffuse adjustment under optimisation frictions. Third, using administrative records of household purchases from the Central Statistics Office, I estimate two buyer-participation difference-in-differences designs to test whether HTB increased the relative importance of first-time buyers within the new-build market.

The paper yields three main findings. First, I document clear excess mass just below the €500,000 cut-off in the treated new-build market during both policy regimes, together with missing mass above it. No comparable pattern appears either in the pre-policy period or in the second-hand market. Under the polynomial counterfactual, the normalised bunching coefficient $\hat{b}$ is 1.352 under the initial regime, and 2.090 under the enhanced regime. This implies a transaction-value response of €6,759 and €10,448 for the marginal developer who relocates below the threshold, around one-third of the maximum statutory relief in each regime. The bunching coefficient rises by 55% against a 50% rise in maximum statutory relief between regimes, in line with a local response of the threshold distortion that strengthens monotonically with subsidy generosity.

Second, the difference-in-differences design shows that this response is concentrated in the bins immediately on either side of the threshold, with limited spillover elsewhere, consistent with Best and Kleven (2018)’s observation that housing-market participants face fewer optimisation frictions than agents in other settings such as labour markets, where frictions can spread the response across a wider range around the cut-off (Anagol et al., 2024).

Third, the buyer-participation response is economically large and precisely estimated. Relative to second-hand purchases, the first-time buyer share of new-build purchases increases by 5.11 percentage points under the initial regime and by 10.50 percentage points under the enhanced regime. Relative to former owner-occupiers, the new-build share of first-time buyer purchases rises by 2.48 and 5.33 percentage points respectively. In both designs, the estimated response scales closely with the doubling of the headline subsidy rate from 5% to 10%, consistent with an approximately linear local buyer-participation response to subsidy generosity over the observed policy range. This participation response was not anticipated when the scheme was designed (Department of Finance, 2022).

This paper makes three contributions. First, it provides the first market-level analysis of Ireland’s HTB scheme, complementing the borrower-level evidence in McCann and Singh (2023) by studying the transaction-value distribution and the cross-segment composition of market activity. Second, it extends the housing-notch literature by examining a partial-coverage notch rather than a universal one. Relative to settings such as Best and Kleven (2018) and Kopczuk and Munroe (2015), this scheme affects only part of market demand, which can attenuate the transaction-value response and create scope for changes in buyer-participation as an empirical object of interest. Third, it adds to a smaller literature on how policy and financing shocks reshape buyer composition in housing markets (Berger, Turner and Zwick, 2020; Carozzi, 2020; Lambie-Hanson, Li and Slonkosky, 2022) by documenting a substantial cross-segment reallocation of first-time buyers from the second-hand market into new-builds under an explicitly targeted policy, and showing that this reallocation scales closely with subsidy generosity.

The remainder of the paper is organised as follows. Section 2 reviews the related literature, and Section 3 describes the policy context. Section 4 develops the theoretical framework, and Section 5 describes the data and presents the associated descriptive statistics. Sections 6, 7, and 8 present the three empirical strategies in turn, each combining method, identifying assumptions, and results. Section 9 concludes.

2 Related Literature

A central result in the housing-tax literature is that non-linear transaction-tax schedules generate excess mass just below salient thresholds. Best and Kleven (2018) establish this result in the context of stamp duty in the United Kingdom, showing substantial bunching below notches and corresponding missing mass above them. Their bunching estimates imply that buyers reduce transaction-values by two to five pounds for each pound of tax avoided at the notch, an amplification driven by leverage amongst downpayment-constrained borrowers. Evidence from other contexts reinforces this core finding. Slemrod, Weber and Shan (2017) also document bunching after transfer-tax reforms in Washington, D.C., with the more salient 2006 reform generating stronger responses than the smaller 2003 reform. Kopczuk and Munroe (2015) similarly show that sales in New York cluster below the mansion-tax threshold, whilst C. K. Y. Leung, T. C. Leung and Tsang (2015) find that both the location and intensity of bunching move when the Hong Kong stamp-duty schedule changes. This evidence suggests that threshold responses depend on the salience, effective magnitude, and position of the notch in the price distribution. It is also consistent with the observation by Best and Kleven (2018) that housing-market participants appear less affected by optimisation frictions than agents in labour markets, responding both quickly to policy changes and rather locally to the cut-off. At the same time, Kopczuk and Munroe (2015) note that housing markets exhibit substantial round-number heaping even in the absence of policy thresholds, a point that is especially important in the present setting as the HTB ceiling lies at €500,000, an inherently salient price point.

Excess mass below a threshold is only part of the distortion. Kopczuk and Munroe (2015) find that the missing mass above the New York threshold exceeds the bunching mass below it, which they interpret as evidence that some mutually beneficial transactions fail to occur because the notch disrupts matching. Evidence on stamp-duty exemptions for first-time buyers further illustrates the broader consequences of such thresholds. Hargaden (2017), studying Ireland’s pre-2007 exemption of this kind, finds a double-bunching pattern in which first-time buyers bunch below the threshold and second-time buyers bunch above it, implying an allocative distortion. Above the threshold, second-time buyers face a smaller tax increase and can therefore outbid first-time buyers more easily. Clemente-

Casinhas and Vale (2026) show that Portugal’s 2024 stamp-duty exemption for first-time buyers under 35 led to a significant increase in asking price growth, with the tax savings offset 3 to 15 months after the policy was implemented. These findings show that threshold-based housing tax policies can generate not only bunching, but also disrupted matching, allocative distortions, and price inflation.

The evidence on price inflation also connects this threshold literature to a related literature on housing purchase assistance more broadly, where the central question is whether such policies improve affordability or are instead capitalised into prices. Krolage (2023), studying a Bavarian housing purchase subsidy, finds price increases for single-family homes consistent with full capitalisation, but no comparable effect for apartments whose buyers were less likely to qualify. Carozzi, Hilber and Yu (2024) show that the incidence of Britain’s Help to Buy equity loan scheme depends on local supply elasticity: in supply-constrained London the policy raised prices without increasing construction, whereas near the English-Welsh border, it increased construction with little price effect. Clemente-Casinhas and Vale (2026) likewise observe that the price response to Portugal’s first-time buyer stamp-duty exemption was strongest in supply-constrained cities and areas with more eligible buyers. This literature therefore shows that local market and supply conditions are central to subsidy incidence.

Eligibility rules may indeed shape the composition of the housing stock itself. Agarwal, Hu and Lee (2023), studying a transaction-tax exemption for new-build homes in New South Wales, Australia, find strong bunching for eligible homes, substantial price capitalisation of the relief, and evidence that eligible homes are smaller and sell more quickly. Uribe (2022) further shows that, under Colombia’s low-cost housing policy, greater subsidy generosity increases bunching at the eligibility threshold and induces households to reduce housing expenditure, which in equilibrium translates into smaller units. Once eligibility depends on a price ceiling, the relevant adjustment margins may extend beyond prices to include housing characteristics and the composition of the stock itself.

This paper also relates to a smaller literature on how policy and financing shocks affect participation patterns in housing markets. Carozzi (2020) finds that the tightening of credit conditions in the United Kingdom disproportionately reduced the volume of transactions made by first-time buyers. Berger, Turner and Zwick (2020) demonstrate that the First-Time Homebuyer Credit in the United States concentrated its effects in the existing-home and starter-home segments rather than in new construction, and that the age distribution of buyers moved towards younger, downpayment-constrained households during the policy period. Lambie-Hanson, Li and Slonkosky (2022) similarly show that, during the housing recovery in the United States, investors gained market share in housing transactions as mortgage credit supply tightened for individual buyers. These papers indicate that policy and financing shocks may reshape not only prices and quantities, but

also which buyers transact and in which market segment.

To my knowledge, the only existing study of Ireland’s HTB scheme is McCann and Singh (2023), who analyse it at the borrower level. They document that the dominant response to the July 2020 enhancement operated through liquidity: eligible first-time buyers reduced their out-of-pocket downpayments by approximately €10,000, the amount of the additional subsidy, with limited effects on leverage or purchase prices. Their borrower-level design, by construction, leaves the market level unexamined: it does not address the €500,000 eligibility ceiling, the distribution of transaction-values around it, or whether HTB drew first-time buyers into the new-build market. This paper addresses these market-level questions.

3 Policy Context

Ireland’s HTB scheme was introduced in the 2017 national budget and has formally applied to qualifying purchases and self-builds from the 1st of January 2017 onwards¹. The scheme provides a cash rebate² funded from the income tax and Deposit Interest Retention Tax (DIRT)³ that the buyer paid over the four tax years preceding the application (Revenue Commissioners, 2024). This makes HTB a direct rebate of the buyer’s own recent tax payments, distinguishing it from the Help to Buy schemes in the United Kingdom, which took the form of government-backed equity or shared-equity loans (DLUHC, 2024; Welsh Government, 2026). The scheme applies only to first-time buyers of new-build properties and is unavailable for properties with a transaction-value above €500,000. Second-hand properties are ineligible, which makes them a natural comparison group in the empirical analysis. Eligibility also requires occupation as a Principal Private Residence⁴, a mortgage with a loan-to-value ratio of at least 70%, and full tax compliance (Revenue Commissioners, 2024).

Let v denote the qualifying transaction-value of the residence, and let T denote the total income tax and DIRT that the buyer has paid over the preceding four years. T therefore acts as an additional ceiling on the relief, alongside the statutory percentage rate and cap.

Under the original regime, which applied from the 1st of January 2017 to the 22nd of July

¹The scheme also briefly allowed retrospective claims for qualifying transactions undertaken between the 19th of July 2016 and the 31st of December 2016. These transactions were completed, and their transaction-values set, before the scheme was announced, so the retrospective eligibility window does not affect the transaction-value data used in the empirical analysis.

²For a purchase, the rebate is paid after the signing of the contract and for a self-build, after the drawdown of the first tranche of the mortgage.

³A tax on interest earned on bank and other deposit accounts in Ireland.

⁴A house or apartment which is owned and occupied as an individual’s only, or main, residence.

2020, relief was given by

$$\text{HTB}_{2017}(v, T) = \begin{cases} \min\{0.05v, \text{€}20,000, T\}, & \text{if } v \leq \text{€}500,000 \\ 0, & \text{if } v > \text{€}500,000. \end{cases}$$

The subsidy therefore equalled 5% of the transaction-value, capped at €20,000, and could not exceed the buyer’s prior four years of eligible tax T . At $v = \text{€}400,000$, the subsidy reaches its maximum, so the marginal subsidy rate falls from 5% to 0%. The original schedule therefore contains a kink at €400,000 and a notch at €500,000, where the subsidy drops to zero.

A temporary enhancement was introduced on the 23rd of July 2020 and has since been extended to the 31st of December 2029. Under this enhanced regime, relief is given by

$$\text{HTB}_{2020}(v, T) = \begin{cases} \min\{0.10v, \text{€}30,000, T\}, & \text{if } v \leq \text{€}500,000 \\ 0, & \text{if } v > \text{€}500,000. \end{cases}$$

Under this version of the scheme, the subsidy is equal to 10% of the transaction-value, capped at €30,000, and again cannot exceed the buyer’s prior eligible tax T . At $v = \text{€}300,000$, the subsidy reaches its maximum, so the marginal subsidy rate falls from 10% to 0%. The enhanced schedule therefore also contains a kink, though at €300,000. At €500,000, the subsidy again drops to zero, creating the same notch in the schedule.

Figure 1 summarises the key non-linearities in the two HTB schedules. The empirical analysis focuses on the common notch⁵, since eligibility is lost entirely at that point and the relevant pricing incentives change discretely.

HTB has operated at substantial scale since its introduction. From 2017 to 2022, approved claims for purchased new-build properties ranged from 3,900 to 5,500 per year (Revenue Commissioners, 2026). At the same time, the new-build market was not composed exclusively of first-time buyers: household former owner-occupiers still accounted for 3,000 to 3,900 such purchases per year over the same period (Central Statistics Office, 2026). Any market-level response at the €500,000 threshold is therefore observed in a segment that includes both HTB-eligible and HTB-ineligible demand.

Over the sample period, rising new-build transaction-values also made the upper end of the eligibility range increasingly relevant. The share of approved claims above €375,000 more than doubled, from 18.4% in 2017 to 43.8% in 2022, tracking a similar rise in the

⁵Although the HTB schedule also contains kinks at €400,000 and €300,000, I do not use a regression kink design at those thresholds, since analogous slope changes are also present in placebo exercises. This limits the credibility of a causal interpretation of the results.

Help to Buy Policy Schedule

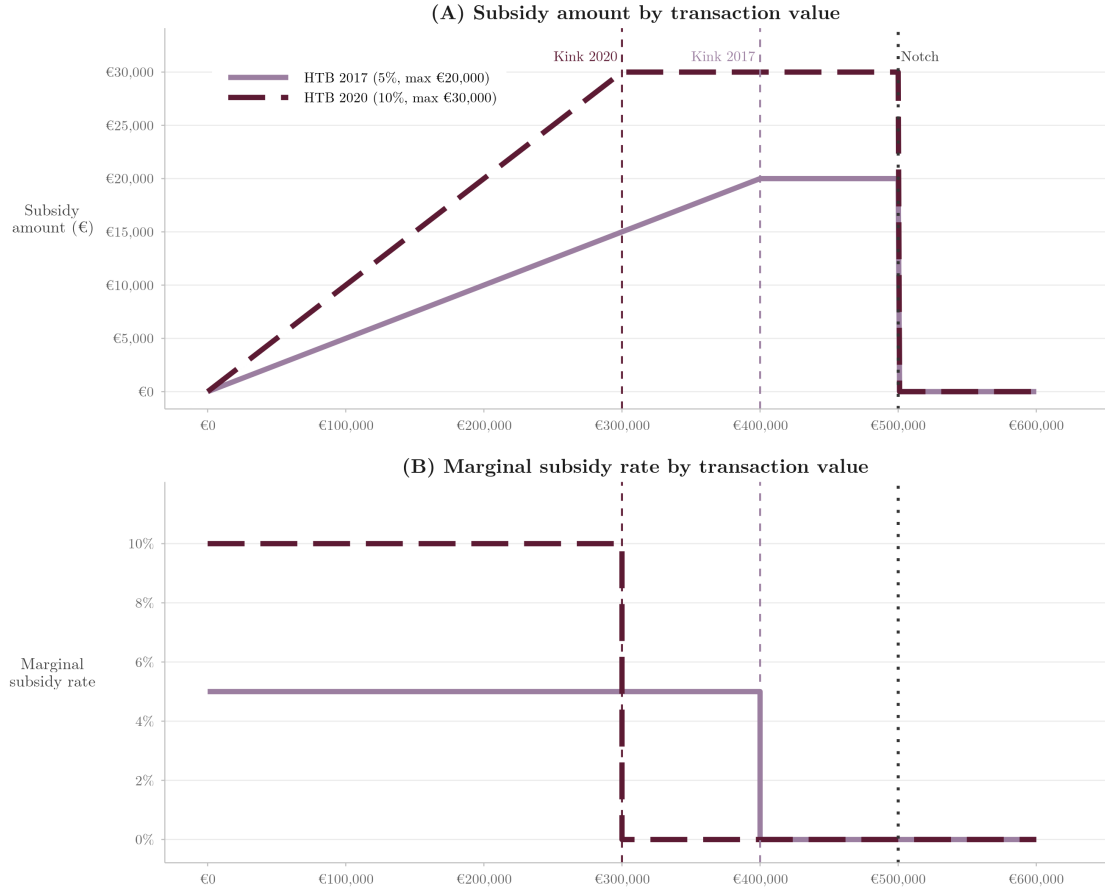


Figure 1: Help to Buy subsidy schedule by qualifying transaction-value.

share of eligible new-build sales above €375,000 from 17.9% to 39.8%⁶. The upward movement in the claim distribution therefore reflects broader new-build price dynamics within the eligibility range rather than a compositional shift specific to HTB claimants. The statutory cap became correspondingly more binding over time. Between 2017 and 2019, 12 to 16% of claims were at the €20,000 maximum. The 2020 enhancement raised the cap to €30,000 but lowered the transaction-value at which it begins to bind from €400,000 to €300,000, exposing a much larger share of new-build transactions to the cap. The share of claims at the new €30,000 maximum rose from 21% in 2020 to 57% by 2022 (Revenue Commissioners, 2026).

The empirical analysis exploits multiple features of this institutional setting: a salient notch at €500,000, a doubling of the headline subsidy rate, a 50% increase in the maximum cap between regimes, and the coexistence of HTB-eligible and HTB-ineligible

⁶Both shares move closely together year by year. For HTB claims and eligible new-build sales respectively: 18.4% and 17.9% in 2017, 20.4% and 25.2% in 2018, 24.2% and 26.7% in 2019, 26.7% and 29.9% in 2020, 29.5% and 31.1% in 2021, and 43.8% and 39.8% in 2022. Source: Revenue Commissioners (2026) and author's calculations from the Property Price Register (Property Services Regulatory Authority, 2026) restricted to new-build transactions at or below €500,000.

demand within the new-build segment.

4 Theoretical Framework

In the Irish new-build market, transaction-values are largely set by developers rather than determined through open bidding⁷. Developers therefore have local discretion at the threshold and are the relevant bunching agents. I accordingly model the developer as choosing the transaction-value of a unit in the presence of a discontinuity in demand at $\bar{v} = \text{€}500,000$: crossing the threshold removes HTB eligibility for purchasers who would otherwise qualify, reducing their willingness to pay and lowering the probability of sale. Developers whose latent preferred transaction-values lie just above $\bar{v}$ may therefore find it more attractive to locate at or just below the cut-off.

4.1 Demand

Following Polyakova and Ryan (2019), I model demand as the aggregation of demands from heterogeneous buyers facing a uniform seller-chosen transaction-value, with targeted subsidies entering through the effective transaction-value paid by eligible consumers. As in Best and Kleven (2018), the analysis is conducted in terms of transaction-values $v = p \cdot h$, where p is the price per unit of housing quality and h is the quality-adjusted quantity of housing services. Since HTB eligibility depends on v rather than on p or h separately, the model treats transaction-value as the choice object: a reduction in v may reflect an adjustment in the unit price, housing characteristics, or both.

Buyers are heterogeneous in willingness to pay and HTB eligibility. Since $v = \bar{v}$ remains eligible, I define demand at the threshold by the left-hand limit, $D(\bar{v}) \equiv D(\bar{v}^-)$. Let $\lambda \in (0, 1)$ denote the effective share of buyers in the relevant new-build segment who are HTB-eligible first-time buyers, $D_I(v)$ the purchase probability of an ineligible buyer at transaction-value v , and $D_E(x)$ the purchase probability of an eligible buyer as a function of their effective transaction-value x . The effective transaction-value satisfies $x = v - \tau$ below the cut-off, and $x = v$ above it. Assume D_I and D_E are continuous and strictly decreasing. The parameter $\tau > 0$ denotes the implicit value of HTB eligibility near $\bar{v}$, interpreted as the reduction in the effective transaction-value faced by an eligible buyer at the threshold.

⁷The Irish Competition and Consumer Protection Commission (2025) notes that, in the case of a new-build home, “the purchase price is usually set in advance and consumers do not engage in a bidding process”.

The resulting reduced-form sale probability faced by the developer is⁸

$$D(v) = \begin{cases} (1 - \lambda) D_I(v) + \lambda D_E(v - \tau) & v \leq \bar{v}, \\ (1 - \lambda) D_I(v) + \lambda D_E(v) & v > \bar{v}. \end{cases} \quad (1)$$

At $\bar{v}$, demand jumps downward by

$$\Delta D \equiv D(\bar{v}^-) - D(\bar{v}^+) = \lambda [D_E(\bar{v} - \tau) - D_E(\bar{v})] > 0, \quad (2)$$

which is positive because D_E is strictly decreasing and $\tau > 0$. I define

$$E(\tau, \lambda) \equiv \lambda [D_E(\bar{v} - \tau) - D_E(\bar{v})], \quad (3)$$

the effective exposed demand at the notch: the demand response induced by losing the subsidy above the cut-off, weighted by the share of demand exposed to that loss.

The partial derivatives of $E(\tau, \lambda)$ make the role of each input explicit:

$$\frac{\partial E(\tau, \lambda)}{\partial \lambda} = D_E(\bar{v} - \tau) - D_E(\bar{v}) > 0, \quad \frac{\partial E(\tau, \lambda)}{\partial \tau} = -\lambda D'_E(\bar{v} - \tau) > 0,$$

where the second expression assumes D_E is differentiable. Effective exposed demand is therefore larger when either the eligible share of demand or the value of eligibility rises.

If, in addition, the policy itself draws more first-time buyers into the new-build market, then λ is endogenous to τ . With $\lambda = \lambda(\tau)$ and $\lambda'(\tau) > 0$, the total derivative of effective exposed demand with respect to subsidy generosity becomes

$$\frac{dE(\tau, \lambda)}{d\tau} = \underbrace{\lambda'(\tau) [D_E(\bar{v} - \tau) - D_E(\bar{v})]}_{\text{participation}} + \underbrace{(-\lambda(\tau) D'_E(\bar{v} - \tau))}_{\text{valuation}} > 0.$$

The participation channel captures that a more generous subsidy increases the eligible share of demand; the valuation channel reflects the fact that each eligible buyer values eligibility more. Both enlarge effective exposed demand at the threshold.

4.2 Developer's Problem

The developer chooses the transaction-value of a single new-build unit. Let $S(v) \in \{0, 1\}$ denote an indicator for whether the unit sells at transaction-value v . Realised profit is

$$\Pi(v) = (v - c(v))S(v),$$

⁸Similar to Polyakova and Ryan (2019), this can be generalised to $D(v) = \int q(v - s(v, \theta), \theta) dF(\theta)$, where θ indexes buyer type and $s(v, \theta)$ denotes the subsidy received by type θ at transaction-value v .

where $c(v)$ is the developer's marginal cost of supplying a unit at transaction-value v . Because the model concerns local decisions in a neighbourhood of the €500,000 threshold, marginal cost can be treated as constant: $c(v) \approx c$. Taking expectations conditional on v yields

$$\pi(v) \equiv \mathbb{E}[\Pi(v) \mid v] = (v - c) \Pr(S(v) = 1 \mid v).$$

Since $D(v)$ denotes the sale probability implied by the demand system, profit can be written as

$$\pi(v) = (v - c)D(v). \quad (4)$$

The developer therefore trades off net revenue per sale, $(v - c)$, against the probability of sale (Guren, 2018). The HTB scheme enters through $D(v)$, and the discontinuity established in (2) implies that, at the cut-off,

$$\pi(\bar{v}^-) = (\bar{v} - c)D(\bar{v}^-), \quad \pi(\bar{v}^+) = (\bar{v} - c)D(\bar{v}^+),$$

so

$$\pi(\bar{v}^-) - \pi(\bar{v}^+) = (\bar{v} - c)[D(\bar{v}^-) - D(\bar{v}^+)] = (\bar{v} - c)E(\tau, \lambda) > 0$$

for any developer with $c < \bar{v}$. Moreover,

$$\begin{aligned} \frac{d}{d\tau} [\pi(\bar{v}^-) - \pi(\bar{v}^+)] &= (\bar{v} - c) \frac{dE(\tau, \lambda)}{d\tau} \\ &= (\bar{v} - c) \left[\lambda'(\tau) (D_E(\bar{v} - \tau) - D_E(\bar{v})) - \lambda(\tau) D'_E(\bar{v} - \tau) \right] > 0. \end{aligned}$$

Thus, the profit gain from locating at or below the threshold is larger when subsidy generosity rises, through both the valuation and participation channels in $dE(\tau, \lambda)/d\tau$.

4.3 Counterfactual Distribution

To connect the individual transaction-value-setting problem to the observed distribution of transaction-values, I allow developers to differ in their local marginal cost c , with smooth density $g(c)$. Following Sallee and Slemrod (2012), I treat the developer's problem as a two-stage decision. In the first stage, long-horizon development and specification choices produce a latent transaction-value v^0 . Close to sale, once v^0 is known, the developer may adjust the final transaction-value locally in response to proximity to $\bar{v}$.

Since developers now differ in c , profit carries an explicit type argument. In the no-notch environment, a developer of type c chooses transaction-value v to maximise

$$\pi^0(v; c) = (v - c)D^0(v),$$

where $D^0(v)$ is the smooth sale-probability schedule that would prevail absent the discon-

tinuity at $\bar{v}$ (Kleven, 2016; Best and Kleven, 2018). Let $v^0(c)$ denote the corresponding optimal transaction-value. If $c \mapsto v^0(c)$ is monotone, it induces a counterfactual density of latent transaction-values

$$f(v) = g((v^0)^{-1}(v)) \left| \frac{d}{dv}(v^0)^{-1}(v) \right|. \quad (5)$$

The key assumption is that $f(v)$ is smooth in a neighbourhood of $\bar{v}$.

4.4 Bunching

Let $\Delta v^* > 0$ denote the upper bound of the latent transaction-value interval for which bunching at $\bar{v}$ is optimal. In the frictionless homogeneous benchmark, the observed transaction-value rule under the notch is

$$v^*(v^0) = \begin{cases} v^0, & v^0 \leq \bar{v}, \\ \bar{v}, & v^0 \in (\bar{v}, \bar{v} + \Delta v^*), \\ v^0, & v^0 > \bar{v} + \Delta v^*. \end{cases} \quad (6)$$

The notch therefore reallocates the counterfactual mass from $(\bar{v}, \bar{v} + \Delta v^*)$ to the single point $\bar{v}$ and leaves transaction-values outside this interval unchanged.

Following Kleven and Waseem (2013), the marginal buncher is the developer type c^* whose latent preferred transaction-value lies exactly at the upper boundary of the bunching region, $v^0(c^*) = \bar{v} + \Delta v^*$, and who is therefore indifferent between bunching at $\bar{v}$ and choosing $v^0(c^*)$:

$$(\bar{v} - c^*)D(\bar{v}) = (\bar{v} + \Delta v^* - c^*)D(\bar{v} + \Delta v^*). \quad (7)$$

4.5 Dominated Region

Because expected profit falls discretely at $\bar{v}$, there exists a neighbourhood above the cut-off in which choosing an interior transaction-value is strictly dominated by bunching at the threshold. For each developer type c with $c < \bar{v}$, the profit gap

$$\pi(\bar{v}; c) - \pi(\bar{v}^+; c) = (\bar{v} - c) E(\tau, \lambda)$$

is strictly positive. By continuity of $\pi(v; c)$ for $v > \bar{v}$, there exists an interval $(\bar{v}, \bar{v} + \Delta v^D(c))$, the dominated region for type c , over which $\pi(\bar{v}; c) > \pi(v; c)$, with the developer indifferent at the upper boundary:

$$(\bar{v} - c)D(\bar{v}) = (\bar{v} + \Delta v^D(c) - c)D(\bar{v} + \Delta v^D(c)).$$

Any developer whose latent transaction-value v^0 falls in this interval therefore locates at $\bar{v}$ rather than above it, and the interval is empty in the observed distribution. Aggregating across developer types yields excess bunching at $\bar{v}$ together with missing mass above the threshold, consistent with standard notch predictions (Kleven and Waseem, 2013; Best and Kleven, 2018). The frictionless benchmark is useful as a reference case, although the empirical response need not take the literal form of a perfect bunching of excess mass at $\bar{v}$ and zero mass immediately above it⁹.

4.6 Implication for Bunching Mass

Following Best and Kleven (2018), consider the homogeneous frictionless benchmark in which there is a single bunching segment $(\bar{v}, \bar{v} + \Delta v^*)$. Let $B(\bar{v})$ denote the excess mass at the threshold, the counterfactual mass reallocated to the cut-off from transaction-values just above $\bar{v}$. With $f(v)$ the counterfactual density absent the notch,

$$B(\bar{v}) = \int_{\bar{v}}^{\bar{v} + \Delta v^*} f(v) dv. \quad (8)$$

If $f(v)$ is approximately flat in a neighbourhood of $\bar{v}$, this simplifies to $B(\bar{v}) \approx f(\bar{v}) \Delta v^*$, which rearranges to give the implied transaction-value response of the marginal buncher that relocates to below the threshold¹⁰:

$$\Delta v^* \approx \frac{B(\bar{v})}{f(\bar{v})}. \quad (9)$$

4.7 Empirical Predictions

Predictions 1. Suppose $E(\tau, \lambda) > 0$ and the counterfactual density $f(v)$ is smooth and strictly positive in a neighbourhood of $\bar{v}$. This theoretical framework delivers four empirical predictions:

- (i) **Excess mass at or just below the notch:** in the HTB-treated new-build market, developers with latent preferred transaction-values just above $\bar{v}$ will relocate to the cut-off, generating excess mass at or immediately below the threshold together with reduced density above it.
- (ii) **Stronger responses under greater subsidy generosity:** the response should be

⁹Developers may only have limited flexibility over transaction-values close to sale, may revise later-phase units as information about demand is revealed, and likely still face untargeted demand from ineligible buyers above the threshold. More generally, notches need not generate zero density above the cut-off under frictions or sparsity in feasible opportunities (Anagol et al., 2024).

¹⁰Equation (9) assumes a single bunching segment and hence a homogeneous population of bunchers. More generally, if developer type θ has transaction-value response $\Delta v(\theta)$ and type-specific counterfactual density $\tilde{f}(v, \theta)$, then $B(\bar{v}) = \int_{\theta} \int_{\bar{v}}^{\bar{v} + \Delta v(\theta)} \tilde{f}(v, \theta) dv d\theta \approx f(\bar{v}) \mathbb{E}[\Delta v]$, so bunching identifies a local average transaction-value response across types near the threshold.

stronger when effective exposed demand $E(\tau, \lambda)$ is larger, which occurs when either the value of eligibility τ or the eligible share of demand λ rises. Empirically, this implies larger distortions under the enhanced regime, which raised the maximum subsidy at the threshold from €20,000 to €30,000.

- (iii) **Positive implied transaction-value response:** the excess bunching mass at the threshold, together with the smooth counterfactual density at $\bar{v}$, identifies the transaction-value response of the marginal buncher,

$$\Delta v^* \approx \frac{B(\bar{v})}{f(\bar{v})},$$

in the homogeneous benchmark, or a local average transaction-value response under heterogeneity.

- (iv) **Buyer-participation:** if HTB increases the relative attractiveness of eligible new-build purchases for first-time buyers, the effective share of eligible demand in the relevant new-build segment, λ , should rise after the policy is introduced.

5 Data and Descriptive Statistics

This section describes the two administrative datasets used in the analysis and how they relate to the paper’s empirical strategy. The Property Price Register is used to study transaction-values and threshold responses around €500,000, whilst the Central Statistics Office series on market-based household purchases is used to analyse buyer-participation in the new-build market. I then present descriptive statistics for both sources.

5.1 Data Description

5.1.1 Property Price Register

The main dataset used in this paper is the Property Price Register (PPR). The PPR is an administrative dataset containing the date of sale, transaction-value, property type, and address of all arm’s length residential property sales in Ireland since the 1st of January 2010 (Property Services Regulatory Authority, 2026). In particular, it records whether a property is classified as a new dwelling or as second-hand, which is central to the empirical strategy developed in this analysis. For new properties, the PPR reports transaction-values exclusive of VAT at 13.5%. Because HTB eligibility and subsidy amounts are defined with reference to the gross transaction-value paid by the purchaser, I adjust new-build transaction-values to be VAT-inclusive throughout the analysis. The register is constructed from stamp-duty filings and, in principle, should therefore capture the near-universe of residential sales.

Some transactions are exempt from stamp duty, or attract it only in reduced form, including certain transfers between spouses or civil partners, transfers following divorce, purchases under the local authority tenant purchase scheme, and specific transfers following certain court-ordered family-property transfers (Citizens Information, 2025). These are unlikely to materially affect the analysis, since they do not constitute standard market transactions. The PPR can therefore be treated as a close approximation to the universe of residential market sales in Ireland.

The estimation sample is restricted to transactions that took place at their full market value¹¹, between the 1st of January 2013 and the 6th of July 2022. This start date excludes the immediate post-crisis period, during which Irish residential construction and new-build transactions remained severely depressed following the financial crisis and recession. These years are less representative of the market environment relevant for the later HTB period. The end date is chosen to avoid contamination from the introduction of the First Home Scheme on the 7th of July 2022, a government-backed shared-equity scheme for new-build purchases that overlaps with HTB and changes the effective set of supports available to first-time buyers.

The analysis is organised around six estimation cells, defined by two property types, new-build and second-hand, as well as three policy regimes. The pre-policy period runs from the 1st of January 2013 to the 31st of December 2016, the initial regime from the 1st of January 2017 to the 22nd of July 2020, and the enhanced regime from the 23rd of July 2020 to the 6th of July 2022.

A limitation of the PPR is that it contains no hedonic property characteristics such as floor area, number of rooms, or building quality. This makes standard hedonic analysis infeasible in the present setting. Rather than attempting to estimate quality-adjusted price effects directly, the empirical strategy in this paper mainly exploits the discontinuous incentives created by the HTB threshold and studies bunching in the transaction-value distribution.

5.1.2 Market-based Household Purchases of Residential Dwellings

To analyse buyer-participation patterns, I use the Central Statistics Office series Market-based Household Purchases of Residential Dwellings (Central Statistics Office, 2026). This is a monthly administrative dataset, also derived from stamp-duty returns, recording the number of executed market purchases of residential dwellings by household buyers¹²

¹¹This excludes transfers such as gifts, related-party sales, and other non-arm's-length transactions that do not reflect market transaction-values.

¹²Here, household buyers refer to private individuals rather than non-household purchasers such as companies, charitable organisations, or state institutions.

by dwelling status, Residential Property Price Index (RPPI) regions¹³, and buyer status. The dwelling-status classification distinguishes new-builds from second-hand dwellings, and buyer type is recorded separately for first-time buyer owner-occupiers, former owner-occupiers, and non-occupier household buyers.

I use the volume-of-sales series based on stamp-duty executions and restrict the sample to a similar period to the PPR, namely January 2013 to the end of June 2022. The series is therefore conceptually close to the PPR in coverage, but it is observed in aggregated rather than transaction-level form. The principal advantage of this series is that it records buyer type, which is not observed in the PPR. Whilst the PPR is used to study transaction-values and bunching around the €500,000 threshold, this data makes it possible to analyse how residential purchases are distributed across treated first-time buyers and untreated former owner-occupiers, as well as other household buyers.

5.2 Descriptive Statistics

5.2.1 Property Price Register

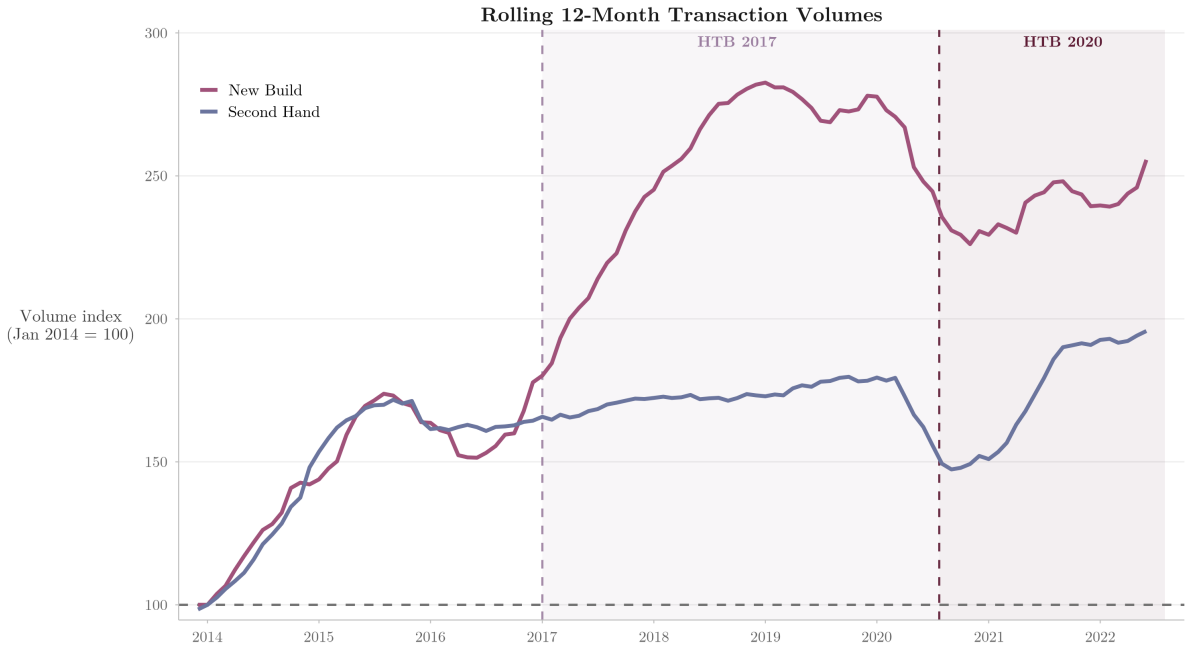


Figure 2: Rolling 12-month transaction volumes for new-build and second-hand properties, with January 2014 normalised to 100. Sample: January 2013 to June 2022.

Figure 2 plots rolling 12-month transaction volumes separately for new-build and second-hand properties. New-build volumes rise much more strongly than second-hand volumes over the sample period, with both series declining in 2020 and 2021 as Covid-19 restrictions disrupted market activity, before both begin to recover in 2022.

¹³The RPPI regional series is slightly finer than the associated county-level series, since it separates the main cities from their surrounding counties for 33 regions in total.

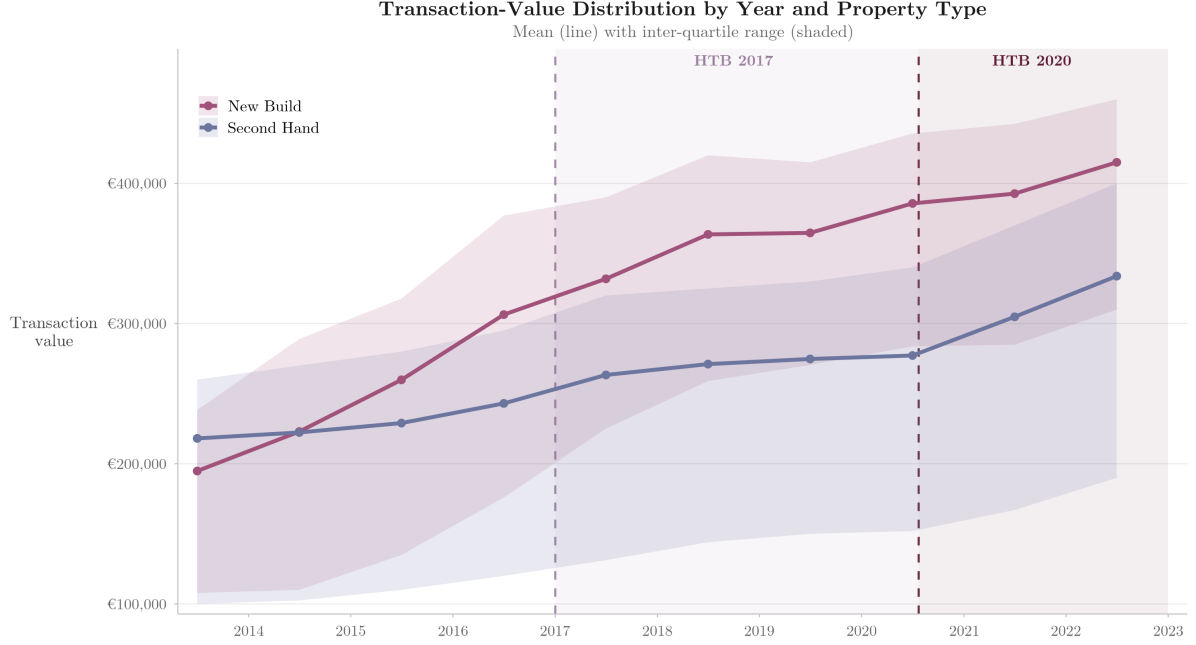


Figure 3: Annual mean and inter-quartile range of transaction-values by property type, 2013–2022. Sample window €50,000–€2,000,000. Sample sizes: pre-policy 20,275 new-build and 131,536 second-hand; initial regime 33,247 and 142,099; enhanced regime 17,971 and 89,289. Full summary statistics in Appendix Table A1.

Transaction-values follow a similar upward trajectory. Figure 3 plots the annual mean and inter-quartile range by property type. Mean new-build transaction-values rise from €255,000 in the pre-policy period to €358,000 under the initial regime and €395,000 under the enhanced regime, based on 20,275, 33,247, and 17,971 new-build transactions respectively. Mean second-hand transaction-values rise more slowly, from €230,000 in the pre-policy period to €270,000 under the initial regime and €304,000 under the enhanced regime, based on 131,536, 142,099, and 89,289 transactions respectively. The inter-quartile range is wide in every cell, indicating that dispersion remains substantial across the full sample, which motivates focusing the bunching analysis on a narrower estimation window around the €500,000 threshold.

I now turn to the transaction-value distributions in the €300,000–€600,000 estimation window. The key descriptive question is whether any clustering at €500,000 can be distinguished from ordinary round-number heaping. Kopczuk and Munroe (2015) note that round-number heaping typically appears in the bin above the salient value rather than below it, so the location of any cluster relative to the cut-off is itself informative.

Figure 4 plots raw histograms in €5,000 bins by regime and property type. Figure 5 reports local polynomial density estimates on the raw data either side of the cut-off, and is used to test whether there exists a discontinuity in the density at €500,000. Appendix Table A2 reports the formal results from this test (Cattaneo, Jansson and Ma, 2018),

computed on the raw data and on a jittered version¹⁴.

Transaction Value Distributions

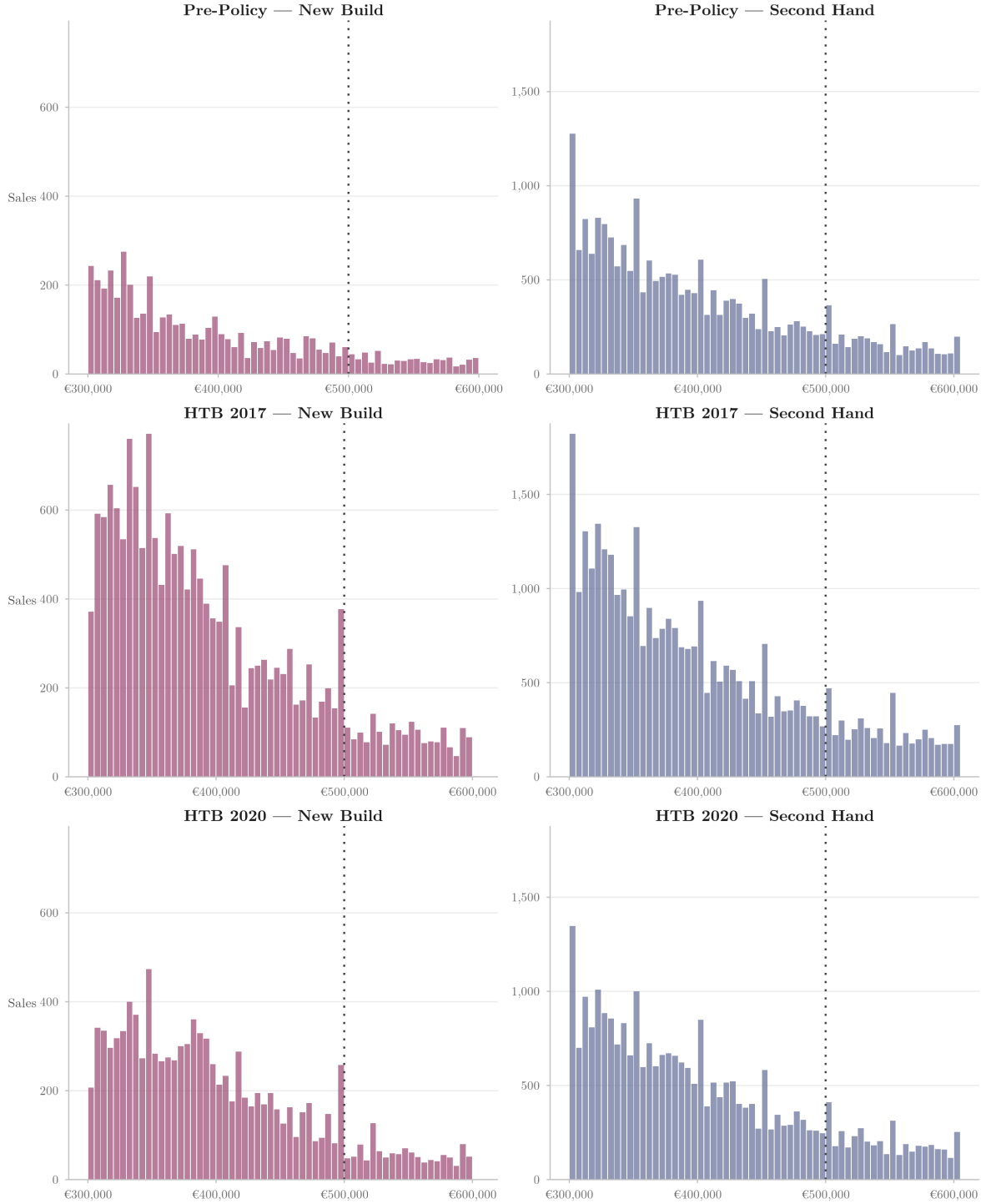


Figure 4: €5,000 bins, by property type and regime. The dotted line marks the €500,000 notch.

¹⁴Each transaction-value is shifted by a uniform draw on $[-€2,500, +€2,500]$. A density break driven by round-number heaping at €500,000 will be dispersed by the jitter and disappear, whilst a genuine discontinuity will not.

Density Discontinuity at the Notch

Local polynomial density estimates ($p = 2$) with robust bias-corrected 95% CIs

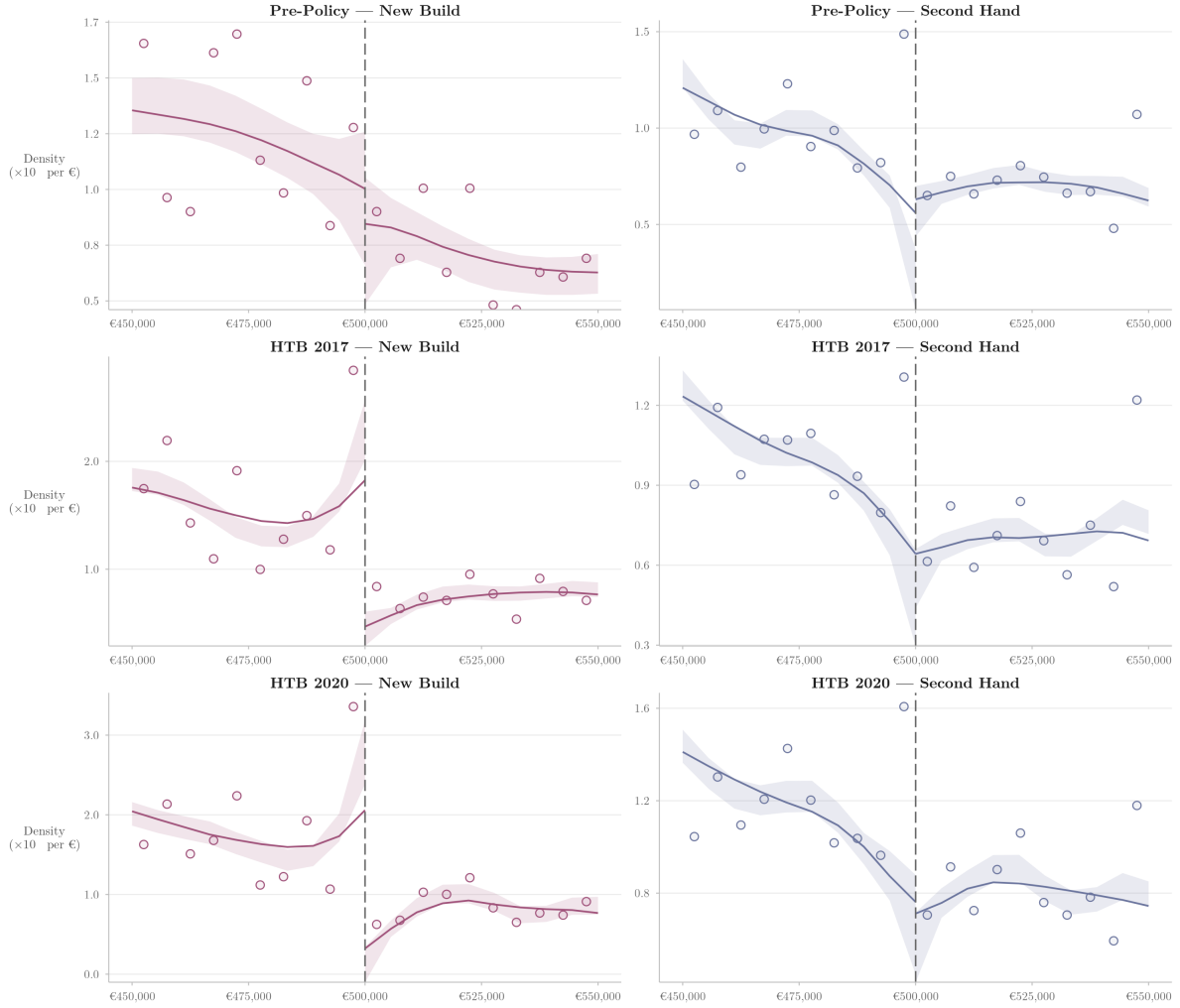


Figure 5: Local polynomial density point estimates and robust bias-corrected 95% confidence intervals. Confidence intervals are robust and bias-corrected, and are therefore not necessarily centred around the point estimate. See Cattaneo, Jansson and Ma (2018) for details.

The new-build panels show a clear policy-response pattern. There is no significant clustering at €500,000 in the pre-policy period, in either the histogram or the density estimates. Under both HTB regimes, transactions cluster clearly in the bin immediately below the cut-off, [€495,000, €500,000), which is the opposite side of the threshold from where pure round-number heaping would appear. Accordingly, the density estimates show a discrete drop at €500,000 with non-overlapping confidence intervals either side. The formal density test in Appendix Table A2 returns large, negative, highly significant breaks under both regimes that remain significant when jittered. The new-build response therefore lies below the threshold and emerges only once HTB is in place, each of which is consistent with a policy-induced response rather than round-number heaping.

The second-hand panels show the opposite pattern and behave as the heaping benchmark predicts. A modest cluster appears in the first bin above the cut-off, [€500,000, €505,000),

in every regime including the pre-policy period, and the density estimates show no break at €500,000 under either HTB regime. The formal density test confirms this: the only significant discontinuity is the pre-policy raw estimate, and it loses significance when jittered. I therefore interpret the second-hand cluster as ordinary round-number heaping rather than a policy response. Second-hand transactions can therefore serve as a comparison group throughout the remainder of the analysis, against which the policy-induced new-build response can be benchmarked.

5.2.2 Market-based Household Purchases of Residential Dwellings

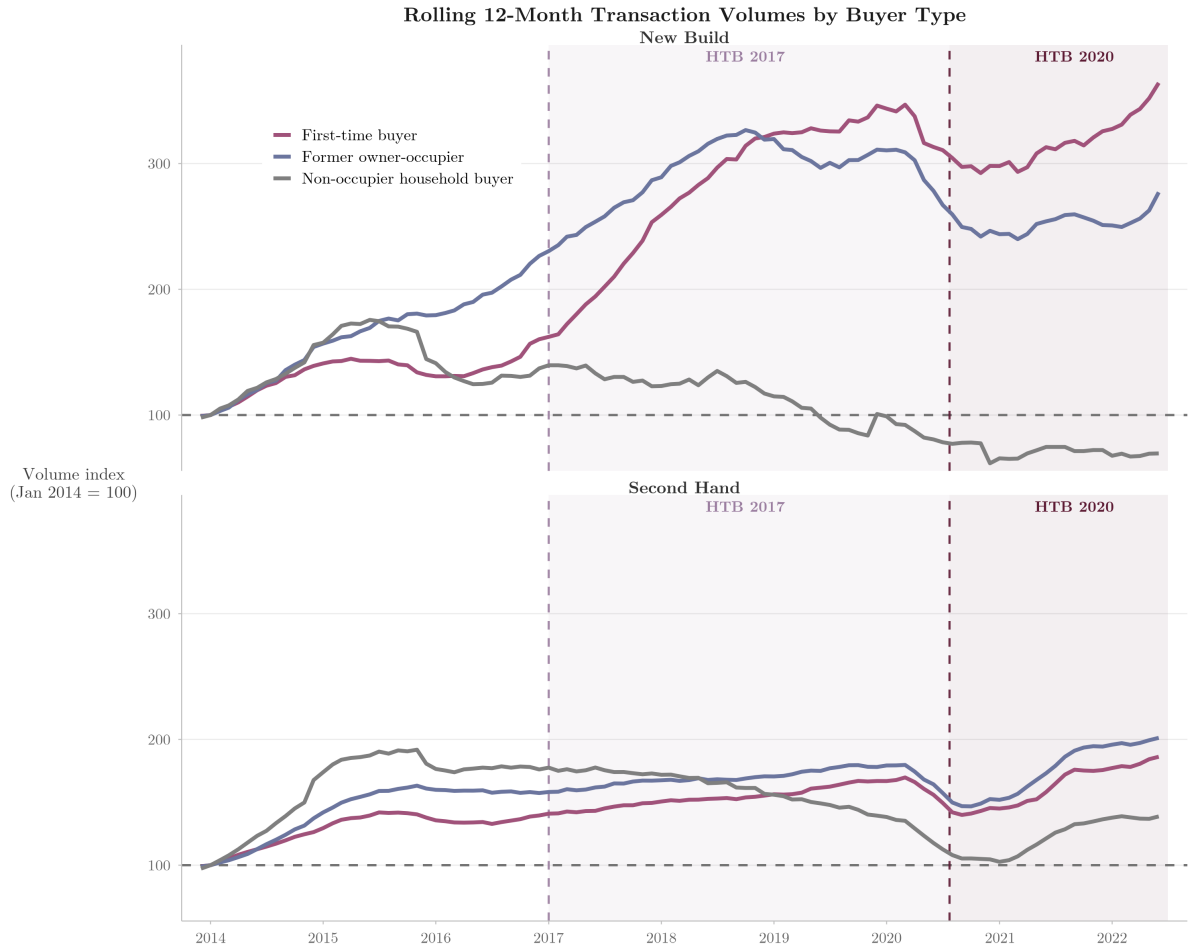


Figure 6: Rolling 12-month transaction volumes by buyer type and dwelling status, with January 2014 normalised to 100. Sample: January 2013 to June 2022.

Figure 6 plots rolling 12-month transaction volumes for first-time buyers, former owner-occupiers, and non-occupier household buyers, separately for new-build and second-hand dwellings. In the new-build segment, all three buyer groups grow during the pre-policy recovery, but their paths diverge over time. Former owner-occupiers grow fastest in the early recovery period, but first-time buyers accelerate after 2017 and overtake them by 2019. Non-occupier household buyers peak around 2015 and then decline steadily, falling

well below their baseline during the enhanced regime. In the second-hand market, the pattern is less differentiated: first-time buyers and former owner-occupiers move broadly in parallel over the sample, whilst non-occupier household buyers again peak earlier and trend downwards during the period of study. The divergence between first-time buyers and other buyer types is therefore concentrated in the new-build segment, which is consistent with a stronger rise in first-time buyer participation in that market.

Table 1: Market-Based Household Purchases by Dwelling Status, Buyer Type, and Regime

		Pre-Policy	HTB 2017	HTB 2020
New Build	First-time buyer	6,733	13,393	8,238
	Former owner-occupier	8,045	12,585	6,221
	Non-occupier household buyer	2,596	1,779	675
Second Hand	First-time buyer	30,482	32,898	20,311
	Former owner-occupier	62,324	66,327	40,856
	Non-occupier household buyer	28,814	24,141	11,682

Table 1 reports descriptive totals from the data by policy regime, dwelling status, and buyer type. Two patterns are clear. First, second-hand purchases greatly exceed new-build purchases in every regime. Second, the buyer mix differs markedly across dwelling status. In the second-hand market, former owner-occupiers are the largest group throughout, accounting for 51.2% of purchases in the pre-policy period, 53.8% under the initial regime, and 56.1% under the enhanced regime. First-time buyers account for 25.1%, 26.7%, and 27.9% respectively, whilst non-occupier household buyers account for 23.7%, 19.6%, and 16.0%.

In the new-build market, by contrast, the split is much more even in the pre-policy period and shifts progressively towards first-time buyers thereafter. First-time buyers account for 38.8% of new-build purchases before the policy, 48.2% under the initial regime, and 54.4% under the enhanced regime. Over the same periods, the share of former owner-occupiers falls from 46.3% to 45.3% and then to 41.1%, whilst the share of non-occupier household buyers declines more markedly from 14.9% to 6.4% and then to 4.5%. These totals therefore suggest that first-time buyers became increasingly prominent within the new-build segment over the policy period. Because the three policy regimes cover different numbers of months, these totals reflect cumulative volumes rather than comparable measures of transaction intensity.

6 Bunching Estimator

6.1 Method

To quantify the extent of threshold responses at €500,000, I estimate the counterfactual transaction-value distribution following the bunching framework of Best and Kleven (2018). The bunching estimator recovers the smooth counterfactual transaction-value distribution that would have prevailed in the absence of the HTB notch and compares it with the observed distribution in a neighbourhood of the cut-off.

Let n_i denote the number of transactions in transaction-value bin i , where transaction-values are grouped into €5,000 bins, and let z_i denote the distance between the lower boundary of bin i and the eligibility threshold $\bar{v}$. I assign transactions priced at exactly €500,000 to $[\text{€}500,000, \text{€}505,000)$, since new-build developers overwhelmingly price just below the threshold rather than at the round-number itself¹⁵.

The counterfactual distribution is estimated by fitting a flexible polynomial to the bin counts, excluding the bins near the threshold that may be distorted by the bunching response, and controlling for systematic clustering at round-number transaction-values:

$$n_i = \sum_{j=0}^q \beta_j (z_i)^j + \sum_{r \in R} \eta_r \mathbf{1} \left\{ \frac{\bar{v} + z_i}{r} \in \mathbb{N} \right\} + \sum_{k=h_{\bar{v}}^-}^{h_{\bar{v}}^+} \gamma_k \mathbf{1} \{i = k\} + e_i, \quad (10)$$

where q is the polynomial order and R is the set of round-number values. The first term is the polynomial in distance from the threshold, which captures the smooth shape of the counterfactual distribution. The second term adds a fixed effect for each bin whose lower boundary is a round-number multiple in R (for example, the bins starting at €450,000 or €525,000), absorbing the heaping that would otherwise be attributed to bunching. The third term adds a separate fixed effect for each bin in the excluded region $(h_{\bar{v}}^-, h_{\bar{v}}^+)$ around the threshold, so that any distortion from bunching does not contaminate the fitted counterfactual. $\mathbf{1}\{\cdot\}$ is the indicator function and $\mathbb{N}$ is the set of natural numbers.

In the baseline specification, I set $q = 5$, use a bin width of €5,000, exclude one bin on either side of the notch, and set $R = \{10,000, 25,000\}$ ¹⁶. The estimation window is restric-

¹⁵Amongst transactions in the $[\text{€}450,000, \text{€}550,000]$ window, 8.2% of new-build transactions (501 sales) are priced in $[\text{€}499,000, \text{€}500,000)$ against 0.03% (2 sales) at exactly €500,000. In the second-hand market, 0.3% (49 sales) are priced in $[\text{€}499,000, \text{€}500,000)$ against 6.2% (1,074 sales) at exactly €500,000. New-build developers therefore avoid the round-number that second-hand sellers gravitate towards, consistent with active threshold avoidance.

¹⁶The set of round-number controls is not identical across all specifications. In an alternative counterfactual exercise in Appendix C, I include controls at €10,000, €25,000, and €50,000, to allow more flexibly for focal-value heaping when the counterfactual is estimated directly from the observed second-hand distribution. In the polynomial bunching estimates, by contrast, the estimation routine permits only two round-number controls, so I use €10,000 and €25,000 as a parsimonious adjustment for focal-value clustering near the notch.

ted to transaction-values between €300,000 and €600,000, which yields 40 bins below the threshold and 20 above it. The lower bound is chosen so that the polynomial counterfactual is fitted over a region with substantial new-build mass, whilst remaining sufficiently tight that the polynomial reflects the local shape of the distribution near €500,000 rather than the broader recovery in lower-value new-build construction. The upper bound reflects the sparsity of new-build transactions above €600,000, since extending the window further would add noise without contributing useful identifying variation.

The counterfactual number of transactions in each bin, $\hat{n}_i$, is obtained by setting the coefficients γ_k on the excluded-region dummies to zero. Excess bunching is defined as the difference between observed and counterfactual bin counts in the excluded bins below the threshold:

$$\hat{B} = \sum_{i \in \mathcal{B}^-} (n_i - \hat{n}_i), \quad (11)$$

where $\mathcal{B}^-$ ends with the final eligible bin [€495,000, €500,000). Missing mass is defined analogously over the excluded bins $\mathcal{B}^+$ at and above the threshold, beginning with [€500,000, €505,000):

$$\hat{M} = \sum_{i \in \mathcal{B}^+} (\hat{n}_i - n_i). \quad (12)$$

I also report the normalised bunching estimate

$$\hat{b} = \frac{\hat{B}}{\hat{n}_0}, \quad (13)$$

where $\hat{n}_0$ is the estimated counterfactual count in the final eligible bin. This normalisation makes bunching estimates comparable across samples with different baseline transaction volumes.

The baseline specification is estimated separately for each of the six regime-by-property-type cells. I report $\hat{B}$, $\hat{M}$, and $\hat{b}$, which map directly to the transaction-value responses in my model, rather than a structural elasticity in the sense of Kleven and Waseem (2013). That elasticity is more natural in tax-notch settings with a well-defined marginal net-of-tax rate, whereas in the present setting a reduction in transaction-value may reflect adjustment in housing characteristics as well as in the posted value itself.

The bunching estimates also identify the implied transaction-value response of the marginal developer who relocates from above the threshold to at or below it. Let $\widehat{\Delta v^*}$ denote the empirical counterpart of Δv^* . Equation (9) establishes that $\Delta v^* \approx B(\bar{v})/f(\bar{v})$. In the discrete-bin setting, the counterfactual density at the threshold is approximated by

$\hat{f}(\bar{v}) = \hat{n}_0/BW$, where BW is the bin width. Substituting into (9) gives

$$\begin{aligned}\widehat{\Delta v^*} &\approx \frac{\hat{B}}{\hat{n}_0/BW} \\ &= \hat{b} \times BW,\end{aligned}\tag{14}$$

so the normalised bunching estimate scaled by the bin width recovers the implied transaction-value response.

Inference is based on the bootstrap distribution of the estimator, and I report percentile bootstrap confidence intervals and bootstrap p -values throughout. An alternative counterfactual strategy that replaces the polynomial benchmark with the second-hand distribution is reported in Appendix C.

6.2 Results

The bunching estimates provide the first direct test of prediction (i) of the model, namely excess mass at or just below the €500,000 notch in the treated new-build market.

Figure 7 plots the observed bin counts against the fitted counterfactual over the €300,000–€600,000 window for each regime-by-property-type cell. Visually, the treated new-build panels show excess mass in the final eligible bin [€495,000, €500,000) and missing mass starting from [€500,000, €505,000), whereas the pre-policy and second-hand panels do not display a comparable pattern beyond ordinary round-number heaping that is present elsewhere in the distribution.

Table 2 formalises this comparison. Under both policy regimes, the treated new-build estimates are positive and statistically significant at the 10% level, though imprecisely estimated, reflecting the smaller scale of the Irish new-build market relative to other contexts. The normalised bunching coefficient is $\hat{b} = 1.352$ under the initial regime and 2.090 under the enhanced regime. By contrast, the pre-policy new-build estimate is $\hat{b} = -0.417$ and statistically indistinguishable from zero, whilst the second-hand estimates are small and statistically insignificant in every period. The threshold response therefore appears only where the policy changes the developer’s problem, in line with prediction (i). Before HTB and in the second-hand market, where $E(\tau, \lambda) = 0$ by construction, pricing above $\bar{v}$ does not generate the same discrete loss in effective demand.

The larger estimate under the enhanced regime is in the direction predicted by the comparative static on τ in Section 4: a larger τ increases the demand loss from pricing above $\bar{v}$, and therefore strengthens the incentive to bunch below the cut-off. The increase in $\hat{b}$ from 1.352 to 2.090, or approximately 55%, is in line with the 50% increase in the maximum subsidy at the threshold from €20,000 to €30,000, consistent with prediction

Bunching Estimator

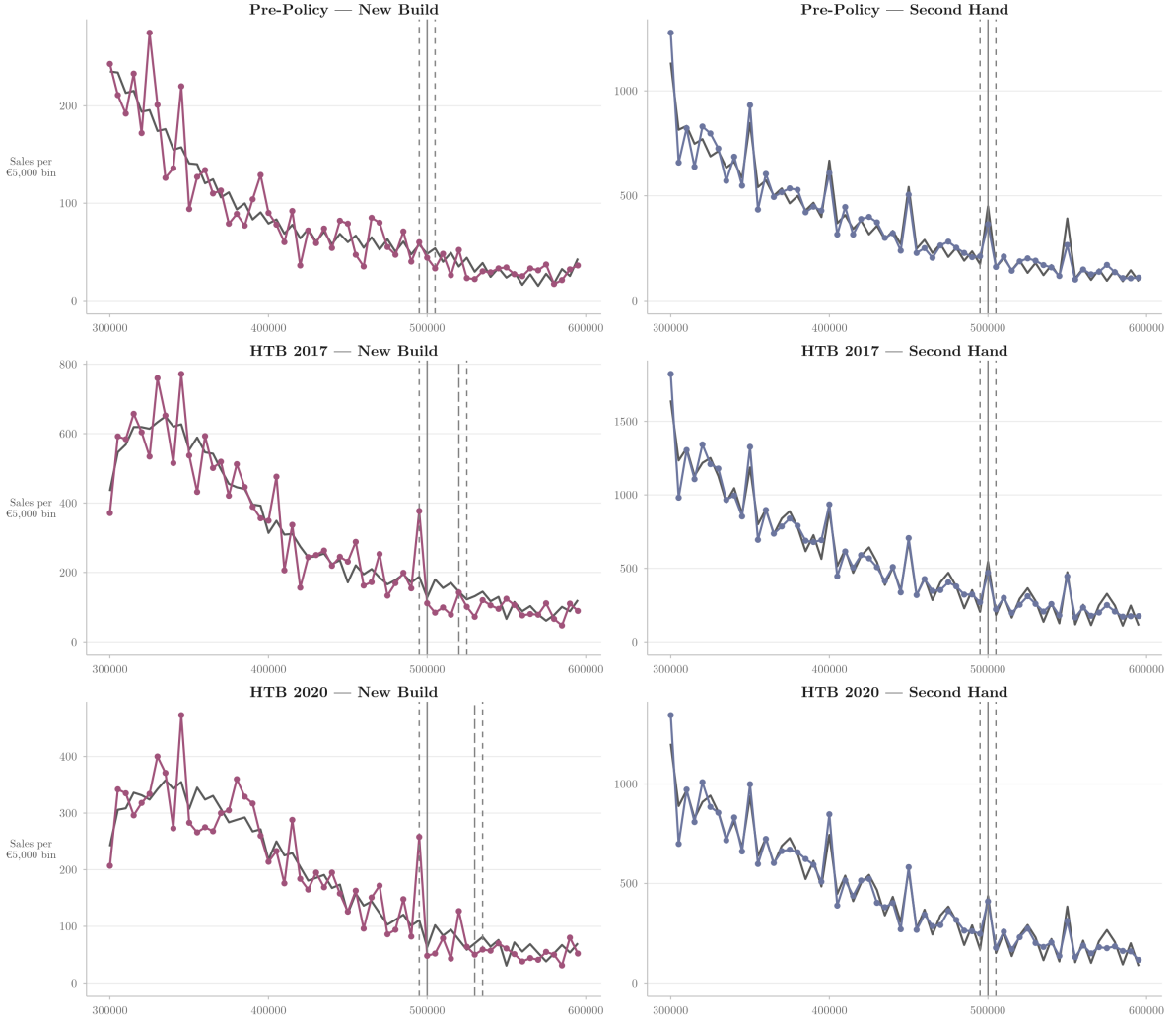


Figure 7: Observed and counterfactual transaction-value distributions around the €500,000 notch, with round-number controls.

(ii) and with the finding in Uribe (2022) that bunching at a housing-subsidy threshold increases monotonically with subsidy generosity. The initial-regime estimate sits somewhat below the 1.64–1.85 range reported by Best and Kleven (2018) at universal stamp-duty notches in the United Kingdom, whilst the enhanced-regime estimate sits just above it. This is consistent with the partial-coverage structure of HTB: only a fraction of demand is exposed to the notch ($\lambda < 1$), which attenuates the response, though a sufficiently large rise in τ may be sufficient to outweigh that attenuation.

The relationship between excess mass below the threshold and missing mass above is itself informative about the form of adjustment. In the frictionless benchmark, mass withdrawn from just above $\bar{v}$ reappears at the cut-off, so one would expect a close correspondence between the two. Empirically, however, the polynomial estimator gives $\hat{B}/\hat{M} = 0.42$ under the initial regime and 0.81 under the enhanced regime, so missing mass exceeds excess mass in both cases. This gap is consistent with two non-exclusive channels. On

Table 2: Bunching Estimates

	Pre-Policy	HTB 2017	HTB 2020
<i>Panel A: New Build</i>			
$\hat{b}$	-0.417 [-2.020, 1.053]	1.352* [-0.229, 4.685]	2.090* [-0.231, 9.343]
$\hat{B}$	-22.1 [-107.2, 55.9]	172.7 [-29.3, 598.4]	132.1 [-14.6, 590.4]
$\hat{M}$	14.5	407.1	162.7
$\hat{B}/\hat{M}$	-1.53	0.42	0.81
<i>Panel B: Second Hand</i>			
$\hat{b}$	-0.177 [-0.873, 0.626]	0.091 [-0.624, 0.886]	0.307 [-0.397, 1.176]
$\hat{B}$	-46.2 [-227.3, 163.0]	28.3 [-193.8, 275.2]	77.4 [-100.1, 296.6]
$\hat{M}$	-61.8	-10.5	17.3
$\hat{B}/\hat{M}$	0.75	-2.70	4.47
Polynomial	5	5	5
RN controls	Yes	Yes	Yes

Notes: Polynomial counterfactual from Eq. (10) with round-number controls at €10,000 and €25,000. 95% percentile bootstrap CIs in brackets (1,000 replications). Bootstrap p -values: * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

the intensive margin, developers may adjust the unit price p or housing characteristics h across a range above the threshold, so that the response is spread over multiple bins rather than concentrated in the final eligible bin. On the extensive margin, some units with latent transaction-values just above the threshold may not be developed at that specification, with developers redirecting resources towards units at other price points, though this channel cannot be separately identified. The increase in $\hat{B}/\hat{M}$ from the initial to the enhanced regime suggests that the adjustment becomes more concentrated near the cut-off as subsidy generosity rises, in line with the model's prediction that a larger τ intensifies the threshold distortion.

Table 3 converts the bunching estimates into implied transaction-value responses using Equation (14). The implied response of the marginal buncher is $\widehat{\Delta v^*} = \text{€}6,759$ under the initial regime and $\text{€}10,448$ under the enhanced regime, corresponding to 33.8% and 34.8% of the maximum statutory relief τ in each regime. These magnitudes are consistent with prediction (iii) of the model. The marginal developer's indifference condition equates the profit gain from preserving eligibility with the profit loss from lowering v , so the implied transaction-value adjustment need not approach the full subsidy. Because $v = p \cdot h$ bundles the unit price with housing characteristics, the response reflects adjustment

in the transacted bundle rather than a pure reduction in the unit price. The ratio of $\widehat{\Delta v^*}$ to τ is around one-third in both regimes, in line with a marginal-buncher response that strengthens with the value of eligibility. The multiple is well below the two-to-five reported by Best and Kleven (2018) at stamp-duty notches in the United Kingdom, where buyers are the bunching agent, consistent with the indirect nature of the developer’s exposure to the notch in this setting: developers face the threshold only through reduced demand from buyers losing eligibility, rather than directly losing the relief themselves.

Table 3: Implied Transaction-Value Response

	HTB 2017	HTB 2020
$\hat{b}$	1.352* [-0.229, 4.685]	2.090* [-0.231, 9.343]
$\widehat{\Delta v^*}$	€6,759 [€-1,147, €23,427]	€10,448 [€-1,156, €46,713]
$\widehat{\Delta v^*}/\tau$	33.8% [-5.7%, 117.1%]	34.8% [-3.9%, 155.7%]
τ	€20,000	€30,000
Bin width	€5,000	€5,000

Notes: 95% percentile bootstrap CIs in brackets (1,000 replications). Bootstrap p -values: * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

6.3 Robustness

I assess the sensitivity of the baseline bunching estimates to four specification choices: the polynomial order q , the bin width, the width of the exclusion zone around the notch, and the number of bootstrap replications. Full results, with detailed discussion, are reported in Appendix B.

The estimates are qualitatively stable across polynomial orders $q \in \{4, 5, 6, 7\}$ and across 500 to 1,500 bootstrap replications, remaining positive and significant in the treated new-build sample under both regimes. The bin-width and exclusion-zone exercises attenuate the estimates because the threshold response operates at a scale narrower than €5,000: as noted in Section 6.1, new-build developers cluster in the [€499,000, €500,000) range, so widening either dimension pools the HTB-sensitive bin with transactions unaffected by the policy. The implied transaction-value response inherits these patterns. Second-hand estimates are close to zero and insignificant across all specifications.

An alternative counterfactual using the second-hand distribution directly is examined in Appendix C, where the bunching response at $\bar{v}$ remains positive and strengthens with subsidy generosity, consistent with the baseline specification.

7 Distributional Difference-in-Differences

7.1 Method

The bunching estimator quantifies excess mass at the €500,000 notch, but does not show how that response is distributed across the wider transaction-value distribution. As emphasised by Anagol et al. (2024), bunching responses need not take the form of a spike at the threshold with zero density immediately above it, but may instead be diffuse in the presence of frictions. I therefore complement the bunching estimates with a stacked distributional difference-in-differences specification that estimates bin-by-bin treatment effects across the full €300,000–€600,000 window, following Cengiz et al. (2019).

I treat the 1st of January 2017 and the 23rd of July 2020 as separate treatment dates and stack them into a common event-time design. For both cohorts, 2014, 2015, and 2016 correspond to $e = -3$, $e = -2$, and $e = -1$ respectively, with $e = -1$ omitted¹⁷. For the initial regime, $e = 0$, $e = 1$, and $e = 2$ correspond to 2017, 2018, and 2019. For the enhanced, $e = 0$ covers the twelve months following the 23rd of July 2020, and $e = 1$ covers the subsequent twelve months, after which the sample ends.

Transactions are grouped into €5,000 bins in the threshold region [€475,000, €525,000), where any bunching or missing-mass response is most likely to appear, and coarser €25,000 bins elsewhere¹⁸. For each bin j , I estimate a linear probability model. Indexing observations by i with transaction-value v_i , and cohorts by $k \in \{I, E\}$ for the initial and enhanced regimes, let NB_i indicate whether observation i is a new-build transaction, $Post_i^k$ equal one if observation i is post-treatment in cohort k , and e_i^k denote event time relative to cohort k 's reform date. Stacking both cohorts gives

$$\mathbf{1}\{v_i \in j\} = \sum_{k \in \{I, E\}} NB_i \cdot \left(\beta_j^k Post_i^k + \gamma_j^k e_i^k + \delta_j^k e_i^k \cdot Post_i^k \right) + \alpha_{cmk} + \varepsilon_{ijcmk}, \quad (15)$$

where α_{cmk} are county-by-year-month-by-cohort fixed effects and standard errors are clustered at the county level. The β_j^k coefficients are the parameters of interest: they capture the discrete jump at cohort k 's reform date in the probability that a new-build transaction falls in bin j , relative to second-hand transactions in the same county, month, and cohort. The γ_j^k terms allow the new-build versus second-hand difference in bin j to evolve linearly with event time, and the δ_j^k terms allow the slope to change at treatment.

¹⁷The year 2013, which would correspond to $e = -4$, is excluded from the stacked design. Unlike the bunching estimator, the distributional difference-in-differences relies on parallel trends between new-build and second-hand transactions over time. Including 2013 would extend the pre-treatment window into a period in which the new-build market was still recovering from the post-crisis collapse in construction, when new-build and second-hand transaction counts evolved very differently.

¹⁸As in the polynomial bunching estimator, transactions priced at exactly €500,000 are assigned to the [€500,000, €505,000) bin rather than to the final eligible bin [€495,000, €500,000), for the reasons discussed in Section 6.1.

Together, they absorb any smooth pre- and post-reform divergence, so that β_j^k isolates the discrete jump at the reform dates. I treat this as the preferred specification, as the descriptive statistics already show pre-existing divergence between the distributions.

7.2 Identifying Assumption

The identifying assumption is that, conditional on the trend and trend-break terms in Equation (15), the probability that a new-build transaction falls in a given bin would have continued to evolve smoothly relative to second-hand transactions through the reform dates in the absence of HTB. To assess this, I estimate a separate event-study specification for each bin j ,

$$\mathbf{1}\{v_i \in j\} = \sum_{e \neq -1} \pi_e^j (NB_i \times \mathbf{1}\{e_i^k = e\}) + \alpha_{cmk} + \varepsilon_{ijcmk}, \quad (16)$$

where $e \in \{-3, -2, 0, 1, 2\}$ with $e = -1$ omitted. For each bin, I combine the two pre-treatment p -values using Fisher’s method¹⁹. The null hypothesis is that the two pre-treatment lead coefficients are jointly equal to zero, implying no differential pre-trend in that bin.

Table 4 indicates that the pre-trends evidence is mixed across the full distribution. Several bins reject the null of no differential pre-trend, particularly in the lower and middle parts of the transaction-value distribution. This is consistent with the descriptive evidence that new-build and second-hand markets were already evolving differently before HTB, and motivates the preferred specification in Equation (15), which does not require that the new-build and second-hand distributions evolve in parallel.

The trend and trend-break terms in Equation (15) are intended to absorb any pre-existing differential evolution between the two markets, so that β_j^k continues to identify the policy effect even when pre-trends are not clean. This absorption is exact only if the pre-treatment trend is linear in event time. Where pre-trends do not reject, no trend needs to be absorbed and β_j^k identifies the policy effect directly. Where pre-trends reject, β_j^k still identifies the policy effect, but under the additional assumption that the underlying differential trend is approximately linear in event time, which these pre-treatment leads cannot directly test.

In the immediate neighbourhood of the policy cut-off, the pre-trends evidence is cleaner above the threshold than below it. The joint Fisher p -value does not reject the null in the final eligible bin or the first ineligible bin, nor in any of the three bins immediately above the threshold. Only at the upper edge of the threshold region, in [€520,000, €525,000), do pre-trends reject. Below the cut-off, by contrast, all three eligible bins in

¹⁹Under the null, $-2 \sum_{e < 0} \log p_e^j \sim \chi_4^2$, where p_e^j is the p -value for $\pi_e^j = 0$.

Table 4: Pre-Trends: Stacked Distributional Event Study

	$t = -3$	$t = -2$	Fisher p
[€300,000, €325,000)	-0.74 (1.77)	8.17 (4.82)	0.255
[€325,000, €350,000)	13.42*** (2.67)	-0.02 (0.95)	0.000
[€350,000, €375,000)	-0.71 (0.80)	-1.50 (0.96)	0.202
[€375,000, €400,000)	-2.38*** (0.65)	-0.22 (2.06)	0.008
[€400,000, €425,000)	-2.71*** (0.96)	-3.13*** (0.89)	0.000
[€425,000, €450,000)	-0.66 (0.49)	0.15 (0.86)	0.461
[€450,000, €475,000)	-3.63*** (1.08)	-0.31 (0.60)	0.011
[€475,000, €480,000)	-0.29 (0.42)	-0.23 (0.20)	0.401
[€480,000, €485,000)	0.39 (0.32)	-0.43** (0.17)	0.023
[€485,000, €490,000)	0.61*** (0.13)	-0.17 (0.29)	0.001
[€490,000, €495,000)	-0.13 (0.11)	-0.29** (0.14)	0.058
[€495,000, €500,000)	-0.27 (0.22)	0.65 (0.39)	0.119
[€500,000, €505,000)	-0.39 (0.36)	-0.19 (0.20)	0.326
[€505,000, €510,000)	-0.25 (0.30)	-0.20 (0.14)	0.251
[€510,000, €515,000)	-0.79** (0.33)	0.02 (0.26)	0.117
[€515,000, €520,000)	-0.12 (0.15)	-0.06 (0.22)	0.717
[€520,000, €525,000)	-0.49*** (0.08)	0.40** (0.16)	0.000
[€525,000, €550,000)	-1.40*** (0.45)	-0.82* (0.46)	0.003
[€550,000, €575,000)	0.28 (0.37)	-0.97*** (0.32)	0.017
[€575,000, €600,000)	1.37*** (0.35)	-0.10 (0.20)	0.003
Observations	131,460		
County $\times$ month $\times$ cohort FE	Yes		
Clustered SEs (county)	Yes		

Notes: * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

$[\text{€}480,000, \text{€}495,000)$ reject the null at conventional levels. The design therefore carries the strongest causal interpretation at the final eligible bin and the first ineligible bin, where the theoretical predictions are located, pre-trends do not reject, and identification does not rely on the trend-break specification.

7.3 Results

I focus on Column (2), which includes cohort-specific linear trends and post-treatment trend breaks for new-build transactions.

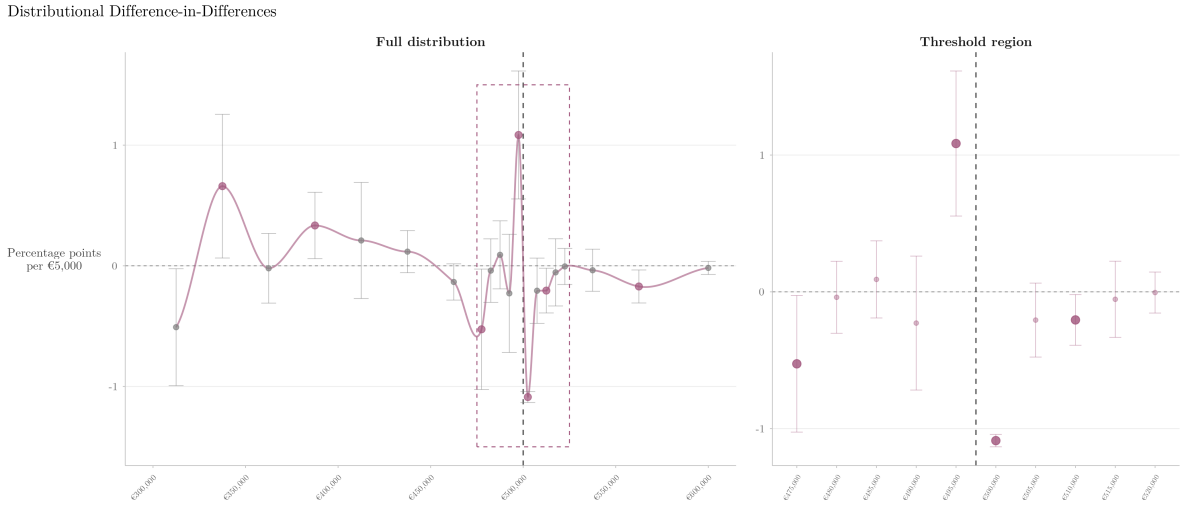


Figure 8: Pooled estimates normalised to percentage points per €5,000.

Table 5 reports the stacked distributional difference-in-differences estimates, whilst Figure 8 summarises the same pattern graphically. The left-hand panel shows a broader upward movement in the new-build transaction-value distribution relative to second-hand, whilst the right-hand panel isolates the clear localised break at $\bar{v}$ by focusing on the threshold region.

Around the threshold, the estimates in Table 5 align with the bunching results. In the final eligible bin, $[\text{€}495,000, \text{€}500,000)$, HTB raised the new-build share of transactions relative to second-hand in the same county-month by 0.630 percentage points under the initial regime and 1.531 percentage points under the enhanced regime, increasing the bin’s probability mass to approximately 1.6 times its pre-policy baseline of 0.98% under the initial regime, and more than two and a half times under the enhanced regime. Both estimates are significant at the 1% level, and the pre-trends diagnostic does not reject differential trends in this bin. In the first ineligible bin, $[\text{€}500,000, \text{€}505,000)$, the corresponding estimate is -0.956 under the initial regime and -1.217 under the enhanced regime, reducing the bin’s probability mass by 65% and 83% relative to the pre-policy baseline of 1.47%, again with clean pre-trends and both estimates significant at the 1%

Table 5: Stacked Distributional Difference-in-Differences

	Mean	(1)		(2)	
		Initial	Enhanced	Initial	Enhanced
[€300,000, €325,000)	19.10%	-3.480*** (0.933)	-4.581** (2.151)	-0.242 (1.181)	-4.806** (2.059)
[€325,000, €350,000)	15.87%	3.259*** (1.108)	1.534 (2.641)	4.341*** (1.107)	2.276 (2.806)
[€350,000, €375,000)	12.71%	1.824** (0.797)	0.243 (1.469)	0.071 (0.751)	-0.275 (1.185)
[€375,000, €400,000)	10.21%	1.369** (0.633)	3.657*** (1.108)	0.759 (0.729)	2.569*** (0.909)
[€400,000, €425,000)	8.77%	-0.095 (0.856)	0.626 (2.094)	0.231 (0.427)	1.856 (2.365)
[€425,000, €450,000)	7.07%	0.359 (0.671)	0.947 (0.570)	0.052 (0.563)	1.113* (0.606)
[€450,000, €475,000)	6.28%	0.273 (0.441)	0.181 (0.553)	-1.247*** (0.348)	-0.101 (0.509)
[€475,000, €480,000)	1.24%	-0.439*** (0.044)	-0.607** (0.262)	-0.492*** (0.084)	-0.559 (0.432)
[€480,000, €485,000)	1.11%	-0.173* (0.092)	-0.310 (0.213)	0.079 (0.243)	-0.156 (0.358)
[€485,000, €490,000)	1.10%	0.170 (0.106)	0.438 (0.323)	-0.078 (0.171)	0.257 (0.204)
[€490,000, €495,000)	0.91%	-0.031 (0.092)	-0.137 (0.205)	-0.178 (0.284)	-0.277 (0.254)
[€495,000, €500,000)	0.98%	1.360*** (0.312)	1.628*** (0.542)	0.630*** (0.146)	1.531*** (0.523)
[€500,000, €505,000)	1.47%	-0.723*** (0.055)	-1.009*** (0.075)	-0.956*** (0.098)	-1.217*** (0.071)
[€505,000, €510,000)	0.70%	-0.206 (0.147)	-0.158 (0.196)	-0.211 (0.198)	-0.201 (0.128)
[€510,000, €515,000)	0.93%	-0.309*** (0.076)	-0.172 (0.118)	-0.123 (0.089)	-0.286 (0.196)
[€515,000, €520,000)	0.59%	-0.126* (0.065)	-0.188 (0.135)	-0.108 (0.087)	-0.001 (0.206)
[€520,000, €525,000)	0.85%	0.040 (0.074)	0.387*** (0.123)	-0.022 (0.076)	0.012 (0.156)
[€525,000, €550,000)	3.41%	-0.983*** (0.206)	-0.528 (0.422)	-0.609 (0.496)	0.234 (0.965)
[€550,000, €575,000)	3.24%	-0.949*** (0.154)	-1.008* (0.559)	-0.989*** (0.249)	-0.726 (0.558)
[€575,000, €600,000)	3.47%	-1.139*** (0.263)	-0.944** (0.437)	-0.907** (0.333)	-1.240*** (0.292)
Observations	131,460				
County $\times$ month $\times$ cohort FE	Yes	Yes	Yes	Yes	Yes
NB $\times$ cohort-specific trend	No	No	Yes	Yes	Yes
NB $\times$ cohort-specific trend break	No	No	Yes	Yes	Yes
Clustered SEs (county)	Yes	Yes	Yes	Yes	Yes

Notes: * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

level. The latter implies a substantial displacement of new-build transactions from the first ineligible bin, relative to where second-hand patterns would otherwise place them. The local sign pattern, with mass rising below the cut-off and falling above, again matches prediction (i). Both estimates strengthen under the enhanced regime, consistent with prediction (ii), but the strengthening is asymmetric: the eligible-bin estimate more than doubles between regimes whilst the ineligible-bin estimate strengthens more modestly, paralleling the rise in $\hat{B}/\hat{M}$ in Section 6.2.

Above the cut-off, the missing mass is very localised: estimates across $[\text{€}505,000, \text{€}520,000)$ are small and statistically insignificant in both regimes, with clean pre-trends. Under optimisation frictions, the response would be expected to spread further from the cut-off (Anagol et al., 2024). Instead, the response is concentrated in the bin immediately above the threshold. The localisation provides some evidence for Best and Kleven (2018)’s suggestion that housing-market participants face fewer optimisation frictions than agents in labour markets. Below the cut-off, the pre-trends evidence complicates interpretation: estimates on the three eligible bins in $[\text{€}480,000, \text{€}495,000)$ are insignificant under the preferred specification, but pre-trends reject in each, so the nulls are equally consistent with a local distortion or with pre-trend contamination. If one is willing to invoke the linearity assumption, the small magnitudes and insignificance in Column (2) would reinforce the localisation reading: across the entire window $[\text{€}480,000, \text{€}550,000)$, the only significant effects appear in the two bins immediately adjacent to the threshold, with no significant effect in any surrounding bin in either regime.

Away from the cut-off, the estimates point to broader relative movements in the new-build distribution that differ by regime. Each regime displays some significant positive and negative coefficients in the lower part of the distribution. These movements are less easily interpreted through the notch mechanism at $\bar{v}$, and may reflect contamination from other features of the HTB schedule, particularly the kinks at $\text{€}400,000$ and $\text{€}300,000$, or the broader upward movement in new-build transaction-values relative to second-hand documented in the summary statistics. The coefficients away from the threshold should therefore be interpreted more cautiously than those in its immediate neighbourhood. As Cengiz et al. (2019) emphasise in a similar distributional difference-in-differences setting, the credibility of this design is strongest in the immediate policy region, where the mechanism operates most directly, rather than further away where confounding trends may dominate. The causal interpretation of these estimates is therefore most direct in the final eligible bin and the first ineligible bin, where the diagnostic in Section 7.2 is clean, and identification does not invoke the additional linearity assumption underlying the trend-break specification.

8 Buyer-Participation

To assess whether HTB also altered who purchases new-build dwellings, I estimate two complementary difference-in-differences designs. The first compares the first-time buyer share across new-build and second-hand dwellings, whilst the second compares the new-build share across first-time buyers and former owner-occupiers. If HTB draws first-time buyers into the new-build segment, both should detect it, and agreement across them would strengthen a policy-driven interpretation. Together, they evaluate prediction (iv) of the model: whether HTB raised the eligible share of demand λ in the new-build segment. As in the distributional difference-in-differences design, the initial and enhanced regime indicators are mutually exclusive, so each interaction coefficient is interpreted relative to the pre-policy period.

8.1 First-Time Buyer Share: New-Build vs Second-Hand

8.1.1 Method

The first design compares the evolution of the first-time buyer share in new-build and second-hand transactions. Let FTB_{rdt} denote the number of first-time buyer purchases in RPPI region r , dwelling-status cell d , and month t , and let HH_{rdt} denote the total number of household purchases in that cell. I define the first-time buyer share as

$$Share_{rdt}^{FTB} = \frac{FTB_{rdt}}{HH_{rdt}} \times 100.$$

I estimate

$$Share_{rdt}^{FTB} = \alpha + \beta_1(NB_d \times Initial_t) + \beta_2(NB_d \times Enhanced_t) + \mu_{rd} + \phi_t + u_{rdt}, \quad (17)$$

where NB_d is an indicator for the new-build segment, $Initial_t$ equals one from the 1st of January 2017 to the 31st of July 2020, and $Enhanced_t$ equals one from the 1st of August 2020 onwards²⁰. The omitted category is the pre-policy period. The region-by-dwelling fixed effects μ_{rd} absorb time-invariant differences between new-build and second-hand markets within each region²¹. The coefficients of interest are β_1 and β_2 ; positive estimates indicate that the first-time buyer share rose more strongly in the new-build segment than in the second-hand segment after the introduction of HTB. Inference is based on p -values from region-clustered standard errors.

²⁰The enhanced regime formally began on the 23rd of July 2020. Because the buyer-participation data is observed at monthly frequency, I code August 2020 as the first month of this policy period.

²¹In the preferred specification, the NB_d main effect is collinear with μ_{rd} , and the $Initial_t$ and $Enhanced_t$ main effects are collinear with the month fixed effects ϕ_t . All three are therefore absorbed by the fixed effects and do not appear in Equation (17).

8.1.2 Identifying Assumption

The identifying assumption is that, absent HTB, the first-time buyer share of new-build transactions would have evolved in parallel with the corresponding share in second-hand transactions²². I assess this using raw annual series, quarterly event-study estimates with the fourth quarter of 2016 as the reference period, and a joint Wald test of the pre-treatment interaction coefficients.

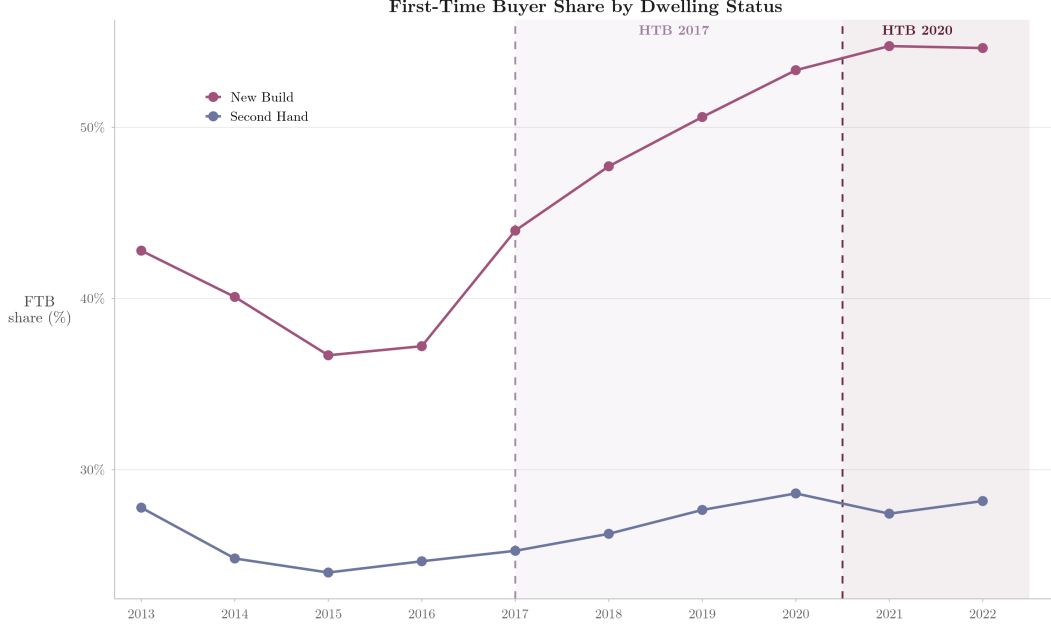


Figure 9: First-time buyer share of transactions by dwelling status, annual.

Figure 9 plots the first-time buyer share separately for new-build and second-hand dwellings. In the pre-policy period, the two series move broadly in parallel, with both shares trending downwards. After the introduction of HTB in 2017, the new-build series rises markedly, whereas the second-hand series increases only slightly.

To examine pre-treatment dynamics more formally, I estimate the following quarterly event-study version of the preferred difference-in-differences specification:

$$Share_{rdt}^{FTB} = \sum_{q \neq 2016Q4} \alpha_q \mathbf{1}\{t \in q\} + \sum_{q \neq 2016Q4} \beta_q (NB_d \times \mathbf{1}\{t \in q\}) + \mu_{rd} + u_{rdt}, \quad (18)$$

where μ_{rd} denotes a fixed effect for each region-by-dwelling cell, and standard errors are clustered by RPPI region. The coefficients β_q trace the evolution of the first-time

²²The Central Bank of Ireland’s macro-prudential mortgage rules were also revised from the 1st of January 2017, coinciding with the introduction of HTB (Central Bank of Ireland, 2016). The main change simplified the first-time buyer loan-to-value limit from a two-tier structure (90% on the first €220,000, 80% above) to a flat 90% on the full property value. This change applied equally to new-build and second-hand purchases, so it differences out in a design that compares dwelling types within this buyer group. The enhanced HTB regime in July 2020 was not accompanied by any change in mortgage lending rules.

buyer share differential between new-build and second-hand transactions over time. The corresponding Wald test evaluates the joint null that all pre-2017 interaction coefficients are equal to zero.

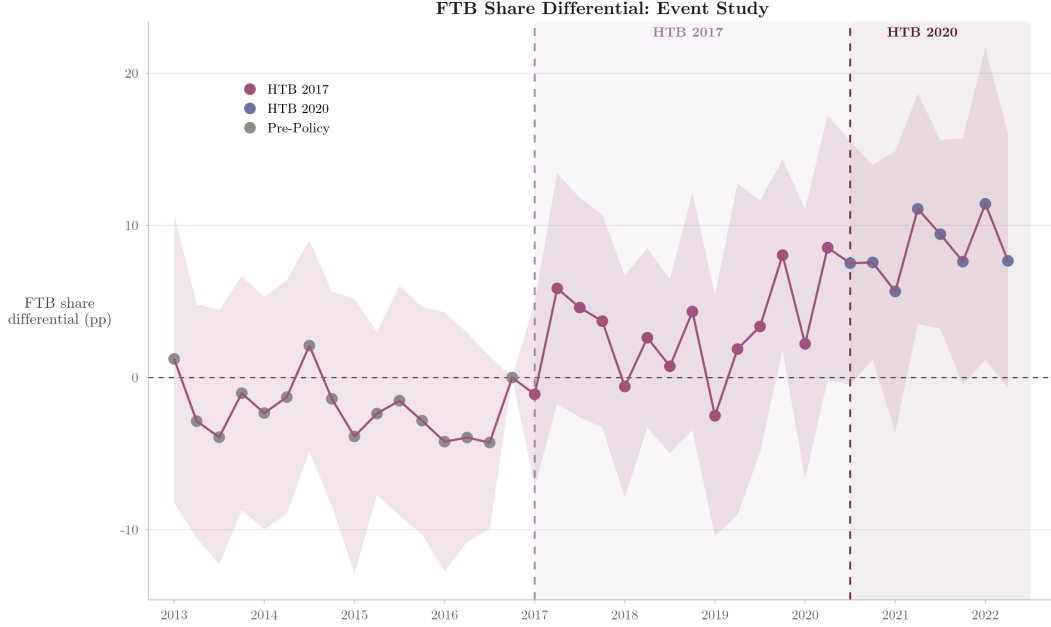


Figure 10: Quarterly event study: first-time buyer share, with 95% confidence intervals. The reference period is the fourth quarter of 2016.

Figure 10 reports the corresponding quarterly event study, with the full set of coefficients in Tables A3 and A4. The pre-treatment coefficients are individually insignificant throughout 2013–2016 (Table A3). The joint Wald test likewise does not reject the null of no pre-treatment differential evolution, with $F(15, 32) = 1.014$, $p = 0.4661$. This non-rejection should not be treated as definitive proof, but in conjunction with the raw series, it is consistent with the identifying assumption used in the main difference-in-differences specification. After 2017, the event-study coefficients are mostly positive and increase further around the enhanced regime, although they are not estimated with a high level of precision. The post-treatment estimates are reported in Table A4.

8.1.3 Results

Table 6 reports estimates of Equation (17). I focus on Column (2), the preferred specification, because the region-by-dwelling fixed effects absorb time-invariant differences between the new-build and second-hand segments within each region. Column (1) yields very similar estimates, so the qualitative conclusion is unchanged.

In the preferred specification, the coefficients on $NB_d \times Initial_t$ and $NB_d \times Enhanced_t$ are 5.11 and 10.50 percentage points respectively, both significant at the 1% level. Relative to the pre-policy first-time buyer share of household purchases of 30.60%, these corres-

Table 6: First-time buyer share of transactions: new build vs second hand

	First-time buyer share (%)	
	(1)	(2)
New Build	13.32*** (1.10)	
New Build $\times$ HTB 2017	5.01*** (1.54)	5.11*** (1.55)
New Build $\times$ HTB 2020	10.55*** (1.93)	10.50*** (1.89)
Dep. var. mean (pre-policy)	30.60	30.60
Observations	7,322	7,322
Adj. R^2	0.291	0.311
Adj. within R^2	0.210	0.013
Region FE	Yes	
Region $\times$ Dwelling FE		Yes
Month FE	Yes	Yes
Clustered SEs	Region	Region
No. regions	33	33

Notes: Each regime coefficient is the full effect relative to the pre-policy period. * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

pond to 16.7% and 34.3% increases in the new-build first-time buyer share beyond any contemporaneous shift in the second-hand market, reallocations the original HTB cost projection had assumed would not occur (Department of Finance, 2022). The enhanced-regime estimate is 2.05 times the initial-regime estimate, closely tracking the doubling of the headline subsidy rate from 5% to 10%. The estimates therefore confirm prediction (iv): λ increased under HTB, and rose more strongly under the more generous regime. Combined with the larger τ under the enhanced regime, this implies a larger $E(\tau, \lambda)$ through both the participation and valuation channels of $dE(\tau, \lambda)/d\tau$, consistent with the stronger bunching response in Section 6.

8.2 New-Build Share: First-Time Buyers vs Former Owner-Occupiers

8.2.1 Method

The second design reverses the comparison. Rather than comparing dwelling types within a buyer group, it compares buyer groups within dwelling types. Let NB_{rbt} and SH_{rbt} denote, respectively, the numbers of new-build and second-hand purchases for buyer group $b \in \{FTB, Former\}$ in region r and month t . I define the new-build share as

$$Share_{rbt}^{NB} = \frac{NB_{rbt}}{NB_{rbt} + SH_{rbt}} \times 100.$$

I estimate

$$\begin{aligned} Share_{rbt}^{NB} = & \alpha + \delta_0 FTB_b + \delta_1 (FTB_b \times Initial_t) \\ & + \delta_2 (FTB_b \times Enhanced_t) + \mu_r + \phi_t + \theta_{rb} t + v_{rbt}, \end{aligned} \quad (19)$$

where FTB_b indicates the first-time buyer group and former owner-occupiers form the comparison group. The region-by-buyer linear time trends $\theta_{rb}t$ allow the new-build share of each buyer group to evolve differently over time within each region²³. The coefficients of interest are δ_1 and δ_2 ; positive estimates indicate that first-time buyers moved towards new-build dwellings more than former owner-occupiers after the introduction of HTB. Inference is again based on p -values from region-clustered standard errors.

8.2.2 Identifying Assumption

The identifying assumption is that, absent HTB, the new-build share of first-time buyer transactions would have evolved in parallel with the corresponding share for former owner-occupiers, conditional on these region-by-buyer linear time trends²⁴. I again assess this using raw annual series, quarterly event-study estimates with the fourth quarter of 2016 as the reference period, and a joint Wald test of the pre-treatment interaction coefficients.

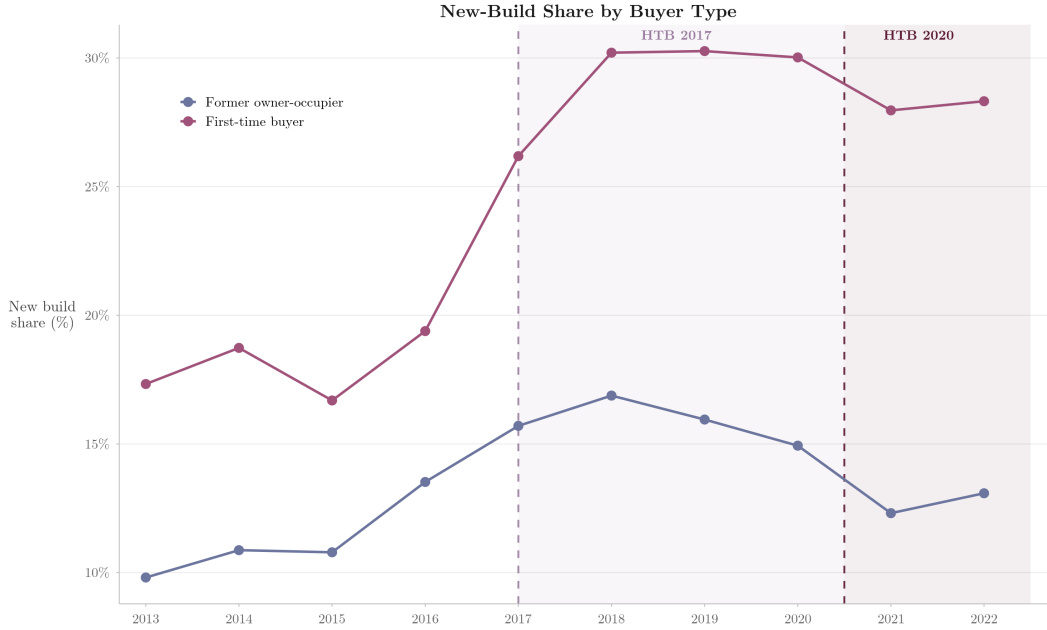


Figure 11: New-build share of transactions by buyer type, annual.

Figure 11 plots the new-build share separately for first-time buyers and former owner-

²³The $Initial_t$ and $Enhanced_t$ main effects are collinear with ϕ_t and are therefore absorbed. The FTB_b main effect δ_0 is identified because the region fixed effects do not vary by buyer type.

²⁴The concurrent 2017 revision of macro-prudential regulations had different impacts on first-time buyers compared with former owner-occupiers. Yet, because this reform applied in the same way to both new-build and existing properties within each buyer category, it does not generate a relative increase in new-build purchases specifically amongst first-time buyers.

occupiers. Both series rise during the pre-policy market recovery, but the gap between them does not display a clear pre-policy widening. After HTB was introduced in 2017, the first-time buyer series increases much more strongly than the former owner-occupier series.

To examine pre-treatment dynamics more formally, I estimate the corresponding quarterly event-study specification:

$$Share_{rbt}^{NB} = \sum_{q \neq 2016Q4} \alpha_q \mathbf{1}\{t \in q\} + \sum_{q \neq 2016Q4} \delta_q (FTB_b \times \mathbf{1}\{t \in q\}) + \mu_{rb} + v_{rbt}, \quad (20)$$

where μ_{rb} denotes a fixed effect for each region-by-buyer-type cell, and standard errors are again clustered by RPPI region. The coefficients δ_q trace the evolution of the differential new-build share between first-time buyers and former owner-occupiers.

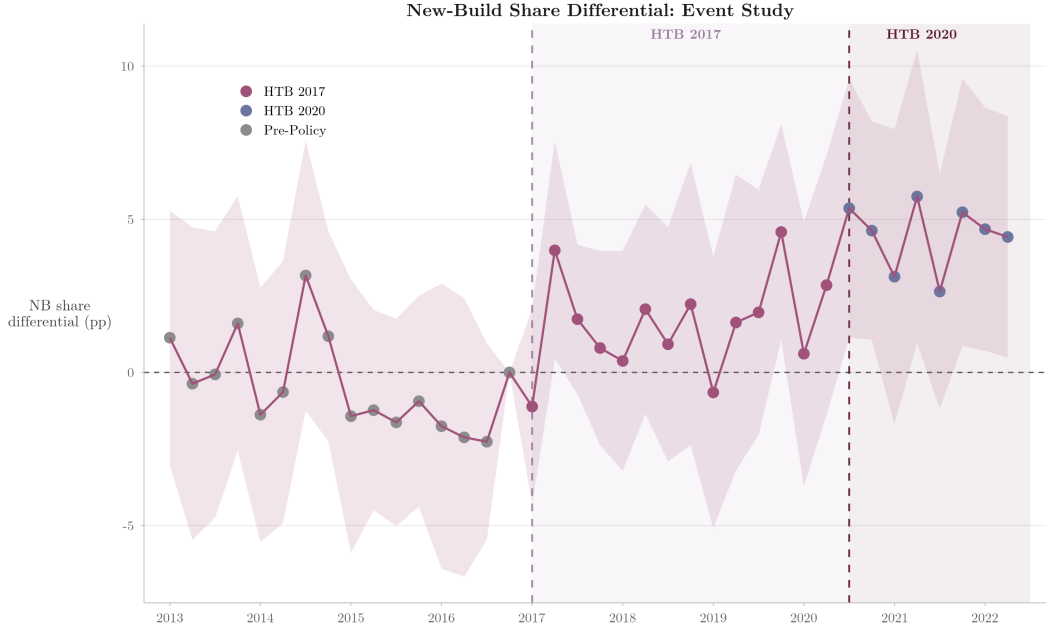


Figure 12: Quarterly event study: new-build share, with 95% confidence intervals. The reference period is the fourth quarter of 2016.

Figure 12 reports the quarterly event study, with the full set of coefficients in Tables A5 and A6. The pre-treatment coefficients are again individually insignificant (Table A5). The joint Wald test does not reject the null of no pre-treatment divergence, with $F(15, 32) = 1.084$, $p = 0.4074$. As with the first design, this does not prove the identifying assumption, but it is consistent with the difference-in-differences interpretation once read together with the raw series. After 2017, the event-study coefficients are mostly positive and increase further around the enhanced regime, although they are not estimated with high precision. The post-treatment estimates are reported in Table A6.

8.2.3 Results

Table 7 reports estimates of Equation (19). As in Section 8.1, I focus on Column (2), which lets the new-build share of first-time buyers and former owner-occupiers evolve differently over time within each region.

Table 7: New build share of transactions: first-time buyers vs former owner-occupiers

	New build share (%)	
	(1)	(2)
First-time buyer	7.05*** (0.67)	7.23*** (0.96)
First-time buyer $\times$ HTB 2017	2.14* (1.09)	2.48*** (0.90)
First-time buyer $\times$ HTB 2020	4.74*** (1.22)	5.33*** (1.43)
Dep. var. mean (pre-policy)	14.22	14.22
Observations	7,523	7,523
Adj. R^2	0.367	0.495
Adj. within R^2	0.145	0.318
Region FE	Yes	Yes
Month FE	Yes	Yes
Region $\times$ Buyer trend		Yes
Clustered SEs	Region	Region
No. regions	33	33

Notes: Each regime coefficient is the full effect relative to the pre-policy period. * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

In the preferred specification, the coefficients on $FTB_b \times Initial_t$ and $FTB_b \times Enhanced_t$ are 2.48 and 5.33 percentage points respectively, both significant at the 1% level. Relative to the pre-policy new-build share of household purchases of 14.22%, these correspond to 17.4% and 37.5% increases in the first-time buyer new-build share beyond the corresponding shift for former owner-occupiers. The enhanced-regime estimate is 2.15 times the initial-regime estimate, close to the 2.05 recovered from the first design. Both responses therefore approximately double between regimes, in line with the doubling of the headline subsidy rate from 5% to 10%, and consistent with an approximately linear local buyer-participation response to subsidy generosity over the observed policy range. Each design therefore identifies the same shift in λ from different sides of the market, and both reinforce the conclusion in Section 8.1 that the buyer-participation channel of $dE(\tau, \lambda)/d\tau$ operates as the model predicts.

The extensive-margin channel of foregone above-threshold units identified in Section 6.2, taken alone, would sit awkwardly with the buyer-participation evidence recovered from

these designs. These two findings are consistent if the segment-level participation response is concentrated in a different part of the transaction-value distribution from the threshold response, and the distributional difference-in-differences in Section 7 supports this reading: relative new-build mass shifts elsewhere in the eligible range rather than solely clustering around the cut-off.

9 Conclusion

This paper has studied how a targeted housing purchase subsidy reshapes behaviour on two margins simultaneously: locally, at the eligibility cut-off, and compositionally, within the treated market. The theoretical framework links those margins through a single object, the share of demand exposed to the notch. Using Ireland’s HTB scheme as its setting, and combining a developer-side notch model with bunching estimation, a distributional difference-in-differences, and two buyer-participation difference-in-differences designs, the analysis shows that both margins respond in tandem to subsidy generosity.

The first margin is the local distortion at the €500,000 eligibility ceiling. Bunching estimation reveals excess mass just below the threshold and missing mass above it in the new-build market during both policy regimes, with no comparable pattern in the second-hand market or in the pre-policy period. Under the polynomial counterfactual, the normalised bunching coefficient $\hat{b}$ is 1.352 under the initial regime and 2.090 under the enhanced regime, implying a transaction-value response of €6,759 and €10,448 for the marginal developer who relocates below the threshold, around one-third of the maximum statutory relief in each regime. Both quantities move with subsidy generosity between regimes: the bunching coefficient is 55% higher against a 50% rise in maximum statutory relief, and the implied transaction-value response grows alongside τ . The distributional difference-in-differences shows that the response is localised at the cut-off rather than diffuse across the wider distribution, and confirms its strengthening under the enhanced regime. Because $v = p \cdot h$, these magnitudes reflect adjustment in the transacted bundle rather than in the unit price alone. The alternative counterfactual in Appendix C preserves the monotonic increase in $\hat{b}$ with subsidy generosity, though the precise cross-regime proportionality is specific to the polynomial specification.

The second margin is the composition of demand in the treated segment. Relative to second-hand purchases, the first-time buyer share of new-build purchases rises by 5.11 percentage points under the initial regime and 10.50 percentage points under the enhanced regime, the latter equivalent to a 34.3% increase over the pre-policy first-time buyer share of household purchases. Relative to former owner-occupiers, the new-build share of first-time buyer purchases rises by 2.48 and 5.33 percentage points respectively. Across both designs, the response scales closely with the doubling of the headline sub-

sidy rate from 5% to 10% that applies to much of the new-build distribution, consistent with an approximately linear local response over the observed policy range. The buyer-participation response therefore moves with subsidy generosity in the same direction as the threshold response, as the theoretical framework predicts when the value of eligibility and the eligible share of demand both rise simultaneously. The buyer-participation response was not anticipated when HTB was designed: the original cost projection assumed first-time buyers would retain the same propensity to purchase new-builds as other buyer types (Department of Finance, 2022). A targeted subsidy therefore does not solely reallocate transactions around its eligibility boundary: it also alters who participates in the treated market, and the framework developed in this paper makes that link explicit.

These findings make three contributions: the first market-level analysis of Ireland’s Help to Buy scheme, complementing the borrower-level evidence in McCann and Singh (2023), an extension of the housing-notch literature (for example, Best and Kleven, 2018; Kopczuk and Munroe, 2015; Agarwal, Hu and Lee, 2023) to the partial-coverage case in which only a subset of demand faces the discontinuity, and evidence on how an explicitly targeted subsidy reshapes buyer composition across market segments (Carozzi, 2020; Berger, Turner and Zwick, 2020; Lambie-Hanson, Li and Slonkosky, 2022).

Several limitations temper these conclusions. The first concerns what the data can measure. The PPR contains no hedonic property characteristics, so the implied transaction-value response cannot be decomposed into adjustments in the unit price p and housing quality h . Settings with richer data, such as those in Agarwal, Hu and Lee (2023) and Uribe (2022), have shown that notches in tax and subsidy schedules can induce developers to deliver smaller units, and whether a similar channel operates in this setting cannot be assessed directly. The same gap leaves open whether bunching at the threshold reflects genuine transaction-value adjustment or is partly explained by unrecorded side payments, a possibility C. K. Y. Leung, T. C. Leung and Tsang (2015) document for stamp-duty notches in Hong Kong. The second limitation concerns the reach of the sample. The estimation window ends in July 2022 to avoid contamination from the First Home Scheme, so any longer-run dynamics, including whether the buyer-participation response attenuates or intensifies under prolonged policy exposure, lie beyond what this analysis can assess.

Three questions therefore remain open. The first is whether the rise in first-time buyer participation this paper documents reflects a real expansion of access to homeownership or instead the acceleration of purchases that would have occurred anyway. For the English HTB scheme, Boileau, Conwell and Levell (2026) find that affordability gains were concentrated amongst higher-income households who could likely have bought without the subsidy, though a shared-equity loan and a tax rebate are sufficiently different instruments that the answer need not be the same.

The second is whether the threshold generates allocative distortions of the kind documented by Hargaden (2017), whose study of Ireland’s pre-2007 stamp-duty exemption finds eligible buyers bunching below the threshold and ineligible buyers bunching above it, with the latter outbidding the former. In the case of new-builds, developers set prices, but an analogous concern operates on two margins. Within the new-build segment, HTB may funnel first-time buyers into units priced just below the cut-off, which may differ in quality, size, or location from those they would otherwise have chosen. Across segments, the buyer-participation evidence shows HTB draws first-time buyers out of the second-hand market and into new-builds, so the second-hand units they would otherwise have acquired transact instead with other buyer groups. The welfare effects of the scheme should therefore be evaluated not only in terms of homeownership access, but also in terms of the dwelling-type margin along which that access shifts. Quantifying these effects, and evaluating counterfactual policy designs, would require a structural model linking the threshold and buyer-participation margins.

The third concerns the incidence of the subsidy. The implied transaction-value response corresponds to approximately one-third of the maximum statutory relief in each regime, well below full capitalisation. Because the absence of hedonic data prevents the decomposition of this response into the unit price and housing quality, the balance between developer capitalisation and quality adjustment cannot be pinned down. Both channels also likely depend on local supply elasticity and the share of eligible buyers in a given area (Carozzi, Hilber and Yu, 2024; Krolage, 2023; Clemente-Casinas and Vale, 2026). Ireland’s segmented housing market, where urban supply constraints are well documented and rural supply is more elastic, is a natural setting for further work on this question. The First Home Scheme, whose eligibility ceilings vary across counties rather than applying at a single national threshold, offers cross-sectional variation in subsidy generosity absent from the single HTB notch. Its interplay with HTB also creates scope for studying how interacting housing subsidies shape buyer behaviour.

For policy, the partial-coverage structure of HTB has implications that universal-coverage notches do not. When every transaction in the market faces the discontinuity, the size of the threshold distortion depends solely on the value of eligibility at the cut-off. When only a subset of demand is exposed, as under HTB, this response also depends on the eligible share of demand in the treated segment. More generous subsidies can therefore strengthen the threshold distortion through two channels: each eligible buyer values eligibility more, and, if participation responds, a larger share of demand becomes sensitive to losing it. The threshold response therefore depends on the full eligibility schedule: the rate and cap determine the value of remaining eligible, whilst the buyer and dwelling restrictions determine how much demand is exposed to the cut-off.

Appendix A Additional Tables

Table A1: Summary Statistics by Regime and Property Type

Regime	Type	Mean	Median	Mode	SD	<i>N</i>
Pre-Policy	New Build	€255,000	€211,000	€147,000	€183,000	20,275
Pre-Policy	Second Hand	€230,000	€175,000	€77,000	€196,000	131,536
HTB 2017	New Build	€358,000	€330,000	€331,000	€180,000	33,247
HTB 2017	Second Hand	€270,000	€220,000	€170,000	€210,000	142,099
HTB 2020	New Build	€395,000	€355,000	€333,000	€199,000	17,971
HTB 2020	Second Hand	€304,000	€250,000	€183,000	€219,000	89,289

Notes: The sample is restricted to sales with a gross transaction-value between €50,000 and €2,000,000 to exclude bulk or portfolio sales and likely non-arm's-length transfers misclassified in the PPR. All transaction-value statistics are rounded to the nearest €1,000.

Table A2: Density Discontinuity Test

	Pre-Policy		HTB 2017		HTB 2020	
	Raw	Jittered	Raw	Jittered	Raw	Jittered
<i>Panel A: New Build</i>						
$\hat{f}^-$	0.96	1.21	2.29	1.98	2.78	2.17
$\hat{f}^+$	0.77	0.90	0.45	0.88	0.12	1.59
T	-0.901	-1.141	-11.274***	-7.209***	-11.458***	-2.097**
p -value	0.368	0.254	0.000	0.000	0.000	0.036
N_{left}	8,915	8,907	24,550	24,462	14,213	14,130
N_{right}	633	641	1,895	1,983	1,153	1,236
<i>Panel B: Second Hand</i>						
$\hat{f}^-$	0.22	0.81	0.48	0.82	0.66	1.19
$\hat{f}^+$	0.57	0.96	0.55	0.78	0.57	1.03
T	3.363***	1.405	0.673	-0.596	-0.738	-1.284
p -value	0.001	0.160	0.501	0.551	0.460	0.199
N_{left}	46,921	47,083	66,851	67,070	47,641	47,818
N_{right}	3,498	3,336	5,121	4,902	4,247	4,070

Notes: Cattaneo, Jansson and Ma (2018). MSE-optimal bandwidth; $p = 2$. Jittered: uniform noise $\pm \text{€}2,500$. * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

Table A3: Pre-Trends: First-Time Buyer Share Event Study

Quarter	NB $\times$ Quarter	p
2013Q1	1.227 (4.823)	0.801
2013Q2	-2.860 (3.917)	0.471
2013Q3	-3.917 (4.259)	0.365
2013Q4	-1.026 (3.921)	0.795
2014Q1	-2.331 (3.895)	0.554
2014Q2	-1.284 (3.917)	0.745
2014Q3	2.104 (3.538)	0.556
2014Q4	-1.393 (3.592)	0.701
2015Q1	-3.865 (4.599)	0.407
2015Q2	-2.361 (2.731)	0.394
2015Q3	-1.518 (3.851)	0.696
2015Q4	-2.831 (3.809)	0.463
2016Q1	-4.210 (4.336)	0.339
2016Q2	-3.936 (3.507)	0.270
2016Q3	-4.272 (2.896)	0.150
Joint Wald $F(15, 32)$	1.014	
p -value	0.4661	
Observations	7,322	
Region $\times$ Dwelling FE	Yes	
Clustered SEs (region)	Yes	

Notes: Pre-treatment coefficients on the quarterly interaction, with standard errors in parentheses. The reference period is the fourth quarter of 2016. The joint Wald test evaluates H_0 : all pre-period interactions = 0. * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

Table A4: Post-Treatment: First-Time Buyer Share Event Study

Quarter	NB $\times$ Quarter	SE
2017Q1	-1.098	(3.089)
2017Q2	5.862	(3.861)
2017Q3	4.600	(3.677)
2017Q4	3.708	(3.566)
2018Q1	-0.588	(3.722)
2018Q2	2.620	(3.001)
2018Q3	0.742	(2.918)
2018Q4	4.335	(3.976)
2019Q1	-2.508	(4.028)
2019Q2	1.877	(5.555)
2019Q3	3.360	(4.219)
2019Q4	8.047**	(3.199)
2020Q1	2.217	(4.523)
2020Q2	8.538*	(4.429)
2020Q3	7.519*	(4.083)
2020Q4	7.566**	(3.264)
2021Q1	5.664	(4.700)
2021Q2	11.086***	(3.862)
2021Q3	9.420***	(3.164)
2021Q4	7.623*	(4.127)
2022Q1	11.417**	(5.225)
2022Q2	7.670*	(4.273)
Observations	7,322	
Region $\times$ Dwelling FE	Yes	
Clustered SEs (region)	Yes	

Notes: Post-treatment coefficients on the quarterly interaction, with clustered standard errors in the final column. The reference period is 2016Q4. * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

Table A5: Pre-Trends: New-Build Share Event Study

Quarter	FTB $\times$ Quarter	p
2013Q1	1.133 (2.119)	0.597
2013Q2	-0.368 (2.605)	0.889
2013Q3	-0.065 (2.385)	0.978
2013Q4	1.600 (2.120)	0.456
2014Q1	-1.381 (2.121)	0.520
2014Q2	-0.642 (2.180)	0.770
2014Q3	3.165 (2.249)	0.169
2014Q4	1.180 (1.747)	0.504
2015Q1	-1.428 (2.274)	0.534
2015Q2	-1.232 (1.670)	0.466
2015Q3	-1.631 (1.724)	0.351
2015Q4	-0.939 (1.757)	0.597
2016Q1	-1.756 (2.376)	0.465
2016Q2	-2.120 (2.318)	0.367
2016Q3	-2.262 (1.635)	0.176
Joint Wald $F(15, 32)$	1.084	
p -value	0.4074	
Observations	7,523	
Region $\times$ Buyer FE	Yes	
Clustered SEs (region)	Yes	

Notes: Pre-treatment coefficients on the quarterly interaction, with standard errors in parentheses. The reference period is the fourth quarter of 2016. The joint Wald test evaluates H_0 : all pre-period interactions = 0. * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

Table A6: Post-Treatment: New-Build Share Event Study

Quarter	FTB $\times$ Quarter	SE
2017Q1	-1.115	(1.630)
2017Q2	3.986**	(1.813)
2017Q3	1.739	(1.240)
2017Q4	0.797	(1.619)
2018Q1	0.375	(1.832)
2018Q2	2.066	(1.743)
2018Q3	0.920	(1.952)
2018Q4	2.229	(2.349)
2019Q1	-0.654	(2.257)
2019Q2	1.629	(2.462)
2019Q3	1.961	(2.044)
2019Q4	4.584**	(1.795)
2020Q1	0.610	(2.197)
2020Q2	2.849	(2.161)
2020Q3	5.361**	(2.153)
2020Q4	4.630**	(1.819)
2021Q1	3.126	(2.458)
2021Q2	5.743**	(2.439)
2021Q3	2.642	(1.950)
2021Q4	5.230**	(2.226)
2022Q1	4.678**	(2.025)
2022Q2	4.423**	(2.010)
Observations	7,523	
Region $\times$ Buyer FE	Yes	
Clustered SEs (region)	Yes	

Notes: Post-treatment coefficients on the quarterly interaction, with clustered standard errors in the final column. The reference period is 2016Q4. * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

Appendix B Bunching Robustness

This appendix reports the full robustness exercises for the bunching estimator, summarised in Section 6.3. The baseline specification fixes the polynomial order at $q = 5$, the bin width at €5,000, the exclusion zone at one bin on each side of the notch, and 1,000 bootstrap replications. This appendix varies each choice in turn.

B.1 Polynomial Order

Table B1: Bunching Estimates: Polynomial Order Sensitivity

		$q = 4$	$q = 5$	$q = 6$	$q = 7$
<i>Panel A: New Build</i>					
HTB 2017	$\hat{b}$	3.048** [0.201, 12.066]	1.352* [-0.229, 4.685]	1.739* [-0.046, 7.879]	1.745* [-0.051, 7.889]
HTB 2020	$\hat{b}$	2.697* [-0.092, 11.831]	2.090* [-0.231, 9.343]	2.183* [-0.383, 13.578]	2.090* [-0.392, 13.225]
<i>Panel B: Second Hand</i>					
HTB 2017	$\hat{b}$	0.065 [-0.635, 0.836]	0.091 [-0.624, 0.886]	0.037 [-0.647, 0.810]	0.045 [-0.662, 0.823]
HTB 2020	$\hat{b}$	0.210 [-0.438, 0.996]	0.307 [-0.397, 1.176]	0.192 [-0.505, 1.032]	0.184 [-0.509, 1.037]

Notes: Baseline specification in bold ($q = 5$). 95% percentile bootstrap CIs in brackets (1,000 replications). Bootstrap p -values: * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

The bunching estimates in Table B1 are qualitatively stable across alternative polynomial orders. In the treated new-build sample, the estimated bunching coefficient is positive and statistically significant at every polynomial order considered under both policy regimes. The $q = 4$ estimate under the initial regime reaches the 5% level, though the magnitude of the effects across regimes is reversed. The baseline and higher-order specifications are significant at 10%. The second-hand estimates remain close to zero and statistically insignificant throughout.

B.2 Bin Width

The bunching estimates in Table B2 are sensitive to the choice of bin width. In the treated new-build sample, the baseline €5,000 specification produces positive and significant estimates in both policy regimes, whereas the €10,000 specification yields smaller and statistically insignificant point estimates²⁵. The micro-level pricing data noted in

²⁵The bin width must divide evenly into the €500,000 threshold for it to fall on a bin boundary. Widths narrower than €5,000 (for example, €2,500) place the threshold in a bin containing too few

Table B2: Bunching Estimates: Bin-Width Sensitivity

		€5k	€10k
<i>Panel A: New Build</i>			
HTB 2017	$\hat{b}$	1.352* [-0.229, 4.685]	0.053 [-0.658, 1.113]
HTB 2020	$\hat{b}$	2.090* [-0.231, 9.343]	0.594 [-0.641, 3.040]
<i>Panel B: Second Hand</i>			
HTB 2017	$\hat{b}$	0.091 [-0.624, 0.886]	-0.251 [-0.652, 0.141]
HTB 2020	$\hat{b}$	0.307 [-0.397, 1.176]	-0.094 [-0.485, 0.323]

Notes: Baseline specification in bold (€5,000 bins). 95% percentile bootstrap CIs in brackets (1,000 replications). Bootstrap p -values: * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

Section 6.1 indicates that developers cluster in the €499,000–€499,999 range, and the distributional difference-in-differences in Section 7 suggests that the response is concentrated in the bins immediately on either side of the threshold. The threshold response therefore operates at a scale narrower than €5,000, and at the €10,000 resolution the fifth-order polynomial cannot isolate this local discontinuity from the broader shape of the distribution, so the bunching mass is absorbed into the smooth fit rather than recovered as excess mass. The €5,000 specification is the coarsest feasible resolution at which the distortion can be recovered cleanly. Second-hand estimates display no comparable sensitivity, sitting close to zero under both bin widths.

Table B3: Implied Transaction-Value Response: Bin-Width Sensitivity

Regime	Bin width	$\hat{b}$	$\widehat{\Delta v^*}$	$\widehat{\Delta v^*}/\tau$
HTB 2017	€5,000	1.352*	€6,759 [€-1,147, €23,427]	33.8% [-5.7%, 117.1%]
	€10,000	0.053	€530 [€-6,578, €11,131]	2.7% [-32.9%, 55.7%]
HTB 2020	€5,000	2.090*	€10,448 [€-1,156, €46,713]	34.8% [-3.9%, 155.7%]
	€10,000	0.594	€5,938 [€-6,414, €30,402]	19.8% [-21.4%, 101.3%]

Notes: 95% percentile bootstrap CIs in brackets (1,000 replications). Bootstrap p -values: * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

transactions for a stable counterfactual and produce degenerate estimates. Widths coarser than €10,000 yield too few bins to the right of the threshold for the fifth-order polynomial to be identified. This leaves €5,000 and €10,000 as the only feasible specifications.

Table B3 shows that the implied transaction-value response inherits the same sensitivity. At the baseline €5,000 bin width, $\widehat{\Delta v}^* = \text{€}6,759$ under the initial regime and €10,448 under the enhanced regime. At a €10,000 bin width, the estimates fall to €530 and €5,938 respectively, and neither is statistically distinguishable from zero.

B.3 Exclusion Zone

Table B4: Bunching Estimates: Exclusion Zone Sensitivity

		1 bin	2 bins	3 bins
<i>Panel A: New Build</i>				
Pre-Policy	$\hat{b}$	-0.417 [-2.020, 1.053]	-0.447 [-2.342, 1.899]	-0.804 [-2.943, 2.223]
HTB 2017	$\hat{b}$	1.352* [-0.229, 4.685]	0.969 [-0.807, 4.423]	0.873 [-1.150, 4.835]
HTB 2020	$\hat{b}$	2.090* [-0.231, 9.343]	1.568 [-0.936, 8.753]	2.335 [-1.057, 15.685]
<i>Panel B: Second Hand</i>				
Pre-Policy	$\hat{b}$	-0.177 [-0.873, 0.626]	-0.307 [-1.211, 0.899]	-0.123 [-1.397, 1.544]
HTB 2017	$\hat{b}$	0.091 [-0.624, 0.886]	-0.078 [-0.902, 0.978]	0.614 [-0.687, 2.493]
HTB 2020	$\hat{b}$	0.307 [-0.397, 1.176]	0.182 [-0.703, 1.383]	0.973 [-0.308, 2.810]

Notes: Baseline specification in bold (1 bin excluded). 95% percentile bootstrap CIs in brackets (1,000 replications). Bootstrap p -values: * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

As emphasised by Kleven (2016), sensitivity to the excluded range also provides a useful diagnostic for bunching estimates. Table B4 compares specifications that exclude one, two, and three bins on each side of the threshold. Point estimates in the treated new-build sample remain positive across all three specifications, though precision weakens substantially as the excluded region widens. Under the initial regime, $\hat{b}$ declines slightly from 1.352 at one excluded bin to 0.969 at two and 0.873 at three, and loses significance once more than one bin is excluded. Under the enhanced regime, the estimate declines at two excluded bins and then rebounds at three, but with a confidence interval wide enough that the response is indistinguishable from zero at both. This pattern reflects the narrow scale at which developers cluster below the notch: widening the exclusion zone removes bins that lie close enough to the cut-off to be informative about the threshold response, rather than absorbing additional bunching mass. The baseline specification with a single excluded bin on each side therefore corresponds to the natural scale of the response rather than an arbitrary choice. In the second-hand sample, by contrast, the

estimates are small and indistinguishable from zero regardless of the excluded range.

B.4 Bootstrap Replications

Table B5: Bunching Estimates: Bootstrap Replication Sensitivity

		$B = 500$	$B = 1,000$	$B = 1,500$
<i>Panel A: New Build</i>				
Pre-Policy	$\hat{b}$	-0.417 [-1.993, 0.927]	-0.417 [-2.020, 1.053]	-0.417 [-1.915, 1.109]
HTB 2017	$\hat{b}$	1.352* [-0.150, 4.600]	1.352* [-0.229, 4.685]	1.352* [-0.185, 4.488]
HTB 2020	$\hat{b}$	2.090* [-0.112, 7.518]	2.090* [-0.231, 9.343]	2.090* [-0.223, 8.684]
<i>Panel B: Second Hand</i>				
Pre-Policy	$\hat{b}$	-0.177 [-0.862, 0.588]	-0.177 [-0.873, 0.626]	-0.177 [-0.891, 0.626]
HTB 2017	$\hat{b}$	0.091 [-0.633, 0.880]	0.091 [-0.624, 0.886]	0.091 [-0.660, 0.954]
HTB 2020	$\hat{b}$	0.307 [-0.373, 1.183]	0.307 [-0.397, 1.176]	0.307 [-0.424, 1.151]

Notes: Baseline specification in bold ($B = 1,000$ bootstrap replications). 95% percentile bootstrap CIs in brackets. Bootstrap p -values: * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

Table B5 shows that the bunching point estimates are invariant to the number of bootstrap replications, as expected, since bootstrap resampling affects inference but not the underlying estimate. Confidence intervals and significance are essentially unchanged from 500 to 1,500 replications in every cell, indicating that the baseline choice of 1,000 replications is sufficient to stabilise inference. Second-hand inference is similarly stable, with estimates close to zero across all three replication counts.

Appendix C Alternative Counterfactual

This appendix reports an alternative bunching exercise that uses second-hand transactions to construct the counterfactual new-build distribution²⁶. The baseline polynomial estimator recovers the untreated distribution from a smooth function fitted to the new-build data itself. That approach is standard, but it necessarily ties the counterfactual to a functional-form approximation of the treated distribution. Here, by contrast, I use an external untreated benchmark. Because second-hand properties were never eligible for HTB, their transaction-value distribution provides a natural reference point for how new-build transactions might have evolved absent the policy.

C.1 Method

To construct the second-hand counterfactual, I model the no-policy new-build distribution as a function of the second-hand distribution and the smooth shape of the transaction-value density. Let $n_i^{NB,0}$ and $n_i^{SH,0}$ denote, respectively, the numbers of new-build and second-hand transactions in bin i during the pre-policy baseline period, where transaction-values are grouped into €5,000 bins²⁷. Let z_i denote the distance between the lower boundary of bin i and the eligibility threshold at $\bar{v} = \text{€}500,000$. I estimate

$$n_i^{NB,0} = \alpha + \delta n_i^{SH,0} + \sum_{j=1}^q \beta_j (z_i)^j + \sum_{r \in R} \eta_r \mathbf{1}\left\{\frac{\bar{v} + z_i}{r} \in \mathbb{N}\right\} + e_i, \quad (\text{C1})$$

where q is the polynomial order and R is the set of round-number values. The intercept α absorbs the average bin count. The second-hand term $\delta n_i^{SH,0}$ captures the extent to which variation in second-hand counts predicts variation in new-build counts across bins. The polynomial $\sum_{j=1}^q \beta_j (z_i)^j$ absorbs the smooth shape of the transaction-value distribution beyond what the second-hand term explains. The round-number fixed effects η_r add a separate dummy for each bin whose lower boundary is a round-number multiple in R , absorbing focal-value heaping that would otherwise be attributed to bunching. $\mathbf{1}\{\cdot\}$ is the indicator function and $\mathbb{N}$ is the set of natural numbers. In this specification, I set $q = 5$ and $R = \{10,000, 25,000, 50,000\}$.

I then construct the treatment-period counterfactual by replacing baseline second-hand

²⁶The logic of this estimator is not dissimilar to Cengiz et al. (2019), who use an untreated group's distribution as the counterfactual in a distributional difference-in-differences framework, rather than relying on a within-group polynomial approximation.

²⁷As in the polynomial bunching estimator, transactions priced at exactly €500,000 are assigned to the [€500,000, €505,000) bin rather than to the final eligible bin [€495,000, €500,000), for the reasons discussed in Section 6.1.

counts with observed second-hand counts from the relevant treatment period:

$$\hat{n}_i^{0,NB}(t) = \hat{\alpha} + \hat{\delta} n_i^{SH}(t) + \sum_{j=1}^q \hat{\beta}_j(z_i)^j + \sum_{r \in R} \hat{\eta}_r \mathbf{1}\left\{\frac{\bar{v} + z_i}{r} \in \mathbb{N}\right\}, \quad (\text{C2})$$

and I rescale $\hat{n}_i^{0,NB}(t)$ so that total predicted new-build transactions match the observed total in treatment period t . Excess bunching $\hat{B}$, missing mass $\hat{M}$, and the normalised bunching coefficient $\hat{b}$ are then computed exactly as in equations (11)–(13), with the second-hand-based counterfactual replacing the polynomial counterfactual. The implied transaction-value response is obtained as $\widehat{\Delta v^*} = \hat{b} \times BW$, as in Equation (14). Inference is based on a bootstrap that resamples underlying transaction-values in both the baseline and treatment periods, and I report percentile bootstrap confidence intervals and bootstrap p -values throughout, as in the polynomial specification.

C.2 Identifying Assumption

The identifying assumption is that, absent HTB, the conditional relationship between new-build and second-hand bin counts would have remained stable over time, after accounting for the polynomial and round-number controls. I assess this with two threshold placebos that re-centre the exercise at €450,000 under each HTB regime, where no policy discontinuity exists. I choose €450,000 because it is a salient round-number in the same part of the transaction-value distribution as the true notch, but remains sufficiently far from the €400,000 kink in the initial regime and the €300,000 kink in the enhanced regime so that the estimate is not contaminated by those slope changes.

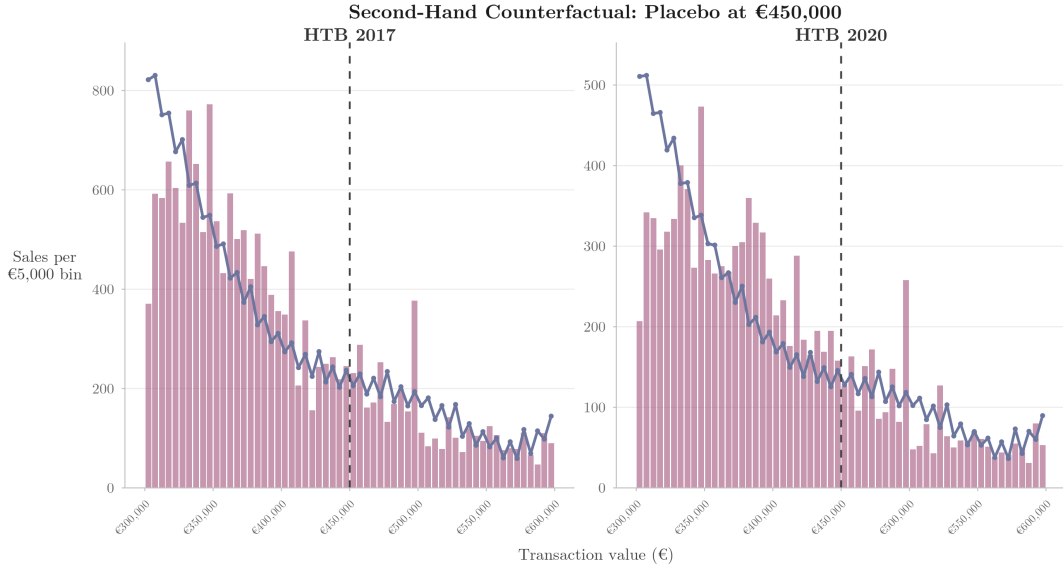


Figure C1: Second-hand-based counterfactual: placebo estimates at €450,000 under each HTB regime. Dashed lines mark the falsified threshold.

Figure C1 plots the observed new-build distribution against the second-hand-based counterfactual in the two threshold placebos. Around the falsified €450,000 threshold, the fit is close, and neither panel displays the pronounced excess mass visible at the true €500,000 notch in the treatment estimates.

Table C1: Second-Hand Counterfactual: Placebo Estimates

	HTB 2017	HTB 2020
$\hat{b}$	0.035 [-0.177, 0.244]	0.085 [-0.120, 0.319]
$\hat{B}$	8.2 [-34.2, 50.7]	12.3 [-15.6, 40.3]
$\hat{M}$	-25.2	1.9
$\hat{B}/\hat{M}$	-0.33	6.35
Polynomial	5	5
RN controls	Yes	Yes

Notes: Counterfactual from Eq. (C1) at €450,000, with round-number controls at €10,000, €25,000 and €50,000. 95% percentile bootstrap CIs in brackets (1,000 replications). Bootstrap p -values:

* $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

Table C1 reports the corresponding estimates. At the falsified €450,000 threshold, the estimated bunching coefficients are 0.035 in the initial regime and 0.085 in the enhanced regime, and neither estimate is statistically significant. The method therefore does not spuriously detect excess mass at a salient round-number threshold where no HTB discontinuity exists. These results are consistent with the null placebo evidence from the main polynomial bunching analysis, despite the different counterfactual constructions, and suggest that the estimated treatment pattern is not an artefact of the method employed.

C.3 Results

Figure C2 plots the observed new-build distribution against the second-hand-based counterfactual for both treatment samples. In each panel, the observed distribution displays clear excess mass in the final eligible bin, [€495,000, €500,000), and missing mass above the threshold extending approximately four bins into the ineligible region, from [€500,000, €505,000) to [€515,000, €520,000). This local pattern is similar to that obtained under the baseline polynomial counterfactual. Around the threshold, the second-hand-based counterfactual passes smoothly through the region and tracks the distribution closely over [€450,000, €550,000), where the bunching response is concentrated.

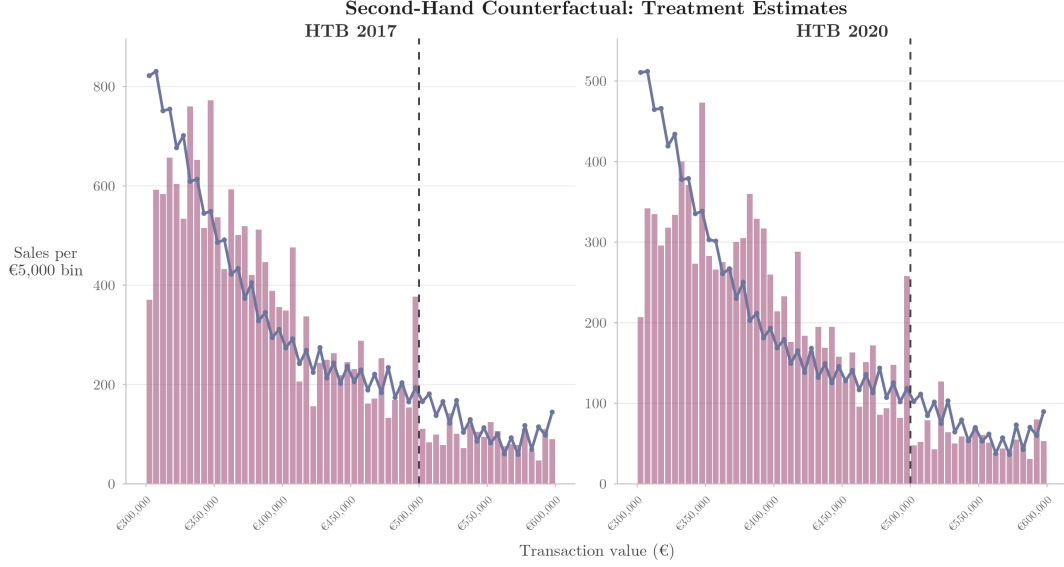


Figure C2: Observed new-build bin counts (bars) and second-hand-based counterfactual (line), by regime. Dashed line marks the €500,000 notch.

Table C2 reports the corresponding estimates. The normalised bunching coefficient is $\hat{b} = 0.947$ under the initial regime and $\hat{b} = 1.178$ under the enhanced regime, both significant at the 1% level. These estimates are smaller than the polynomial-counterfactual estimates in Table 2 (1.352 and 2.090), particularly under the enhanced regime, but they again confirm prediction (i) of the model: substantial excess mass below the threshold, with a stronger normalised response under the more generous regime, in line with prediction (ii) and the comparative static on τ in Equation (2). A further notable feature is that the second-hand-based confidence intervals exclude zero under both regimes, whereas the polynomial intervals straddle it, so the second-hand counterfactual delivers materially tighter inference at the cost of somewhat smaller point estimates.

Table C2: Second-Hand Counterfactual: Treatment Estimates

	HTB 2017	HTB 2020
$\hat{b}$	0.947*** [0.792, 1.486]	1.178*** [1.002, 1.805]
$\hat{B}$	183.4 [142.0, 224.7]	139.5 [106.9, 171.9]
$\hat{M}$	54.7	54.2
$\hat{B}/\hat{M}$	3.35	2.57
Polynomial	5	5
RN controls	Yes	Yes

Notes: Counterfactual from Eq. (C1) with round-number controls at €10,000, €25,000 and €50,000.

95% percentile bootstrap CIs in brackets (1,000 replications). Bootstrap p -values: * $p < 0.10$;

** $p < 0.05$; *** $p < 0.01$.

The cross-regime scaling of the bunching coefficient is, however, somewhat weaker under the second-hand counterfactual than under the polynomial specification. Under the polynomial counterfactual, $\hat{b}$ rises by approximately 55% between regimes against a 50% increase in the maximum statutory relief. Under the second-hand counterfactual, the corresponding increase is from 0.947 to 1.178, or about 24%. Both specifications therefore show that the bunching response strengthens with τ , in line with prediction (ii), but the near-proportionality between $\hat{b}$ and τ recovered from the polynomial counterfactual does not survive the alternative counterfactual.

This attenuation operates through the denominator of $\hat{b}$. The excess-mass estimate $\hat{B}$ declines by approximately a quarter between regimes under both specifications, from 172.7 to 132.1 under the polynomial and from 183.4 to 139.5 under the second-hand counterfactual. The two estimators differ instead in how the counterfactual notch density $\hat{n}_0$ moves between regimes. The polynomial counterfactual recovers $\hat{n}_0$ from a smooth function fitted only to new-build data, which is relatively stable across regimes. The second-hand counterfactual, by contrast, lets $\hat{n}_0$ inherit movement in the second-hand distribution at the threshold through δ , so $\hat{b} = \hat{B}/\hat{n}_0$ is sensitive to common drift in the surrounding density in a way the polynomial estimator is not. The direction of the comparative static on τ is preserved, but the proportional scaling is less robust to the choice of counterfactual than the baseline specification alone would suggest.

The second-hand counterfactual also places these estimates more uniformly within a partial-coverage reading of the broader housing-notch literature. Under the polynomial counterfactual, the initial-regime $\hat{b} = 1.352$ lies below the 1.64–1.85 range reported by Best and Kleven (2018), whereas the enhanced-regime $\hat{b} = 2.090$ exceeds it. Under the second-hand counterfactual, both regime estimates lie below the Best and Kleven (2018) range. This pattern gives a cleaner illustration of the $\lambda < 1$ attenuation logic developed in Section 4: when demand exposed to the notch is only a fraction of total new-build demand, the bunching response at the cut-off is mechanically dampened relative to universal-notch benchmarks. This result holds across both HTB regimes with the second-hand counterfactual, rather than only under the initial schedule with the polynomial counterfactual.

One significant difference relative to the polynomial counterfactual lies in the estimated missing mass above the threshold and, relatedly, in the implied $\hat{B}/\hat{M}$ ratio. Under the polynomial counterfactual in Table 2, missing mass exceeds excess mass in both regimes ($\hat{B}/\hat{M} = 0.42$ and 0.81), whereas under the second-hand counterfactual the reverse is true: $\hat{B}/\hat{M} = 3.35$ under the initial regime and 2.57 under the enhanced regime. The excess-mass estimates below the threshold are broadly similar across the two approaches. The divergence comes primarily from how the counterfactual is fitted in the region above the notch. This suggests that the precise decomposition of the response into excess mass

below and missing mass above is sensitive to the choice of counterfactual, even though the existence of substantial bunching below €500,000 is not.

The direction of the $\hat{B}/\hat{M}$ change is again informative. Under the second-hand counterfactual, the observed pattern moves closer to that reported by Agarwal, Hu and Lee (2023) for New South Wales, where excess mass exceeding missing mass is interpreted as evidence of net entry into the subsidised segment. The polynomial results are not consistent with a pure net-entry interpretation, but the buyer-participation evidence in Section 8 shows that HTB does draw additional first-time buyers into the new-build segment, raising the effective share of demand exposed to the notch, λ . The second-hand counterfactual therefore produces a mass decomposition that aligns more naturally with the participation channel identified by the buyer-participation designs than the polynomial specification does, even though the first-stage existence of threshold bunching is robust across both constructions.

Table C3 converts these estimates into implied transaction-value responses. The implied response is $\widehat{\Delta v^*} = \text{€}4,735$ under the initial regime and $\widehat{\Delta v^*} = \text{€}5,890$ under the enhanced regime, corresponding to 23.7% and 19.6% of the maximum statutory relief respectively.

Table C3: Second-Hand Counterfactual: Implied Transaction-Value Response

	HTB 2017	HTB 2020
$\hat{b}$	0.947*** [0.792, 1.486]	1.178*** [1.002, 1.805]
$\widehat{\Delta v^*}$	€4,735 [€3,960, €7,430]	€5,890 [€5,010, €9,025]
$\widehat{\Delta v^*}/\tau$	23.7% [19.8%, 37.1%]	19.6% [16.7%, 30.1%]
τ	€20,000	€30,000
Bin width	€5,000	€5,000

Notes: Second-hand counterfactual from Eq. (C1). 95% percentile bootstrap CIs in brackets (1,000 replications). Bootstrap p -values: * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

Relative to the polynomial-counterfactual estimates in Table 3, the second-hand counterfactual responses are smaller and more precisely estimated, in line with the smaller bunching coefficients and tighter confidence intervals reported above. The implied transaction-value response remains positive under both regimes, confirming prediction (iii), and is larger under the more generous schedule, confirming the direction of the comparative static on τ in prediction (ii). As in the baseline specification, the implied adjustment remains well below the full subsidy, which is consistent with the marginal

buncher's indifference condition in Equation (7). Again, since $v = p \cdot h$, the estimate should be interpreted as adjustment in the transacted bundle rather than as a pure reduction in the unit price.

A notable difference relative to the baseline specification concerns the ratio $\widehat{\Delta v^*}/\tau$. Under the polynomial counterfactual, this ratio is around one-third in both regimes, which Section 6.2 reads as evidence that the marginal buncher's response strengthens with the value of eligibility. Under the second-hand counterfactual, the ratio falls from 23.7% to 19.6% as τ rises, so the response still grows in absolute terms, but does not scale proportionally. As with the cross-regime pattern in the bunching coefficient itself, the near-exact proportionality that the baseline specification delivers is therefore partly a feature of the polynomial counterfactual rather than a robust property of the data. The direction of prediction (ii) is preserved, but the proportionality of the response to τ is not itself robust to the choice of counterfactual.

Table C4 examines the sensitivity of the second-hand counterfactual transaction-value response to the choice of bin width. As in the polynomial bunching estimator, I report the €5,000 baseline alongside a coarser €10,000 specification²⁸.

Table C4: Second-Hand Counterfactual: Implied Transaction-Value Response (Bin-Width Sensitivity)

Regime	Bin width	$\hat{b}$	$\widehat{\Delta v^*}$	$\widehat{\Delta v^*}/\tau$
HTB 2017	€5,000	0.947***	€4,735 [€3,960, €7,430]	23.7% [19.8%, 37.1%]
	€10,000	0.461***	€4,610 [€3,120, €8,360]	23.1% [15.6%, 41.8%]
HTB 2020	€5,000	1.178***	€5,890 [€5,010, €9,025]	19.6% [16.7%, 30.1%]
	€10,000	0.496***	€4,960 [€3,850, €9,110]	16.5% [12.8%, 30.4%]

Notes: Second-hand counterfactual from Eq. (C1). 95% percentile bootstrap CIs in brackets (1,000 replications). Bootstrap p -values: * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

The response remains positive and significant at the 1% level under both bin widths, and the enhanced-regime estimate exceeds the initial-regime estimate throughout. This

²⁸The same constraint as in the polynomial estimator applies: bin widths must divide evenly into the €500,000 threshold for it to fall on a bin boundary, and widths coarser than €10,000 leave too few bins above the threshold for the polynomial component to be identified. The narrower €2,500 specification, although technically feasible here since the second-hand counterfactual does not require the iterative bunching-region procedure used by the polynomial estimator, is unstable in this setting: with only a few bunching transactions per bin at that resolution, the bootstrap distribution of $\hat{b}$ is heavily skewed and the percentile interval excludes the point estimate. I therefore restrict the comparison to €5,000 and €10,000, the same range reported in Table B3.

contrasts markedly with the polynomial counterfactual, where the €10,000 estimates in Table B3 fall to €530 and €5,938 and lose statistical significance entirely. Under the second-hand counterfactual, the €10,000 estimates are €4,610 and €4,960 respectively, and remain significant at the 1% level. The improvement in bin-width robustness reflects the greater structure imposed by the external benchmark above the threshold, where the polynomial specification is most dependent on functional-form extrapolation and where, in the theoretical framework of Section 4, the dominated region above $\bar{v}$ is located.

The second-hand counterfactual reproduces the core findings of the paper using an external untreated benchmark in place of the within-sample polynomial. It passes its own placebo test, detecting no excess mass at the falsified €450,000 threshold under either regime. It again confirms predictions (i) to (iii): substantial excess mass below the €500,000 notch, a stronger normalised response under the enhanced regime, and a positive implied transaction-value response. In two respects, it yields stronger results than the polynomial baseline: its confidence intervals exclude zero under both regimes, where the polynomial intervals straddle it, and its implied transaction-value response is far less sensitive to bin width. However, some aspects of the polynomial results are not replicated under the second-hand counterfactual: the near-proportional scaling of $\hat{b}$ and $\widehat{\Delta v^*}$ with τ across regimes, and the precise decomposition of the response into excess mass below and missing mass above the threshold, both depend on how the counterfactual is fitted in the region above the notch, and are not robust to the change of counterfactual. The direction in which the second-hand counterfactual moves that decomposition, with excess mass exceeding missing mass, is nonetheless consistent with the buyer-participation channel of Section 8. I therefore place greater weight on the sign and significance of $\hat{b}$ and $\widehat{\Delta v^*}$, which hold across both counterfactual constructions, than on the cross-regime scaling factor or the $\hat{B}/\hat{M}$ ratio, which do not.

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